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Computer Application in Health Care II Final 206 1

Health records and information technology (Kenya Medical Training College)

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Computer Application in Health Care II

Code CAH 206

Module Competence

This module is designed to enable the learners set up a computer, carry out basic maintenance and apply appropriate computer application packages in health records management and communication

Module Outcomes

By the end of this module, the learner should be able to;

a) Carry out basic hardware connectivity and maintenance
b) Understand and explain the computer application software
c) Use Ms. Ward for daily activities
d) Apply Ms. Excel in Health Records Management
e) Use PowerPoint for data presentation
f) Browse internet and communicate by email

Module Units

Unit Name	Hours	
	Theory	Practical
➢ Basic Hardware Connectivity and Maintenance	2	2
➢ Computer Applications Packages	2	0
➢ Ms. Word	6	16
➢ Ms. Excel	6	16
➢ Ms. Power Point	2	4
➢ Internet and Email	2	2

Module Contents

1. **Basic Hardware Connectivity and Maintenance;**

Computer Ports, PS/ 2, Serial Port, Parallel Ports, Audio Ports, Video Ports, VGA Port, Digital Video Interface (DVI), Display Port, HDMI,USB, Connecting a computer to power and computer devices, Booting a computer, How BIOS work, Boot devices , Boot sequences.

2. **Computer Applications Packages;**

Word processor (Microsoft Word, WordPerfect, Lotus Word pro, Work Pages)

Database Processing, (ADABAS, IBM DBS2, Microsoft Access, Microsoft SQL Server, MySQL,Oracle RDBMS, QuickBase)

Spreadsheet calculation (Openoffice.org calc, Microsoft Office Excel, GoogleDocs, Gnumeric, Apple iWork Numbers, Lotus 1-2-3, Libreoffice Calc)

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3. **Microsoft Office Word (Ms. Word)**

Creating a new document, Opening a document, Previewing, saving and closing a document

Using help, Exiting word, entering and deleting text, selecting and replacing text, Navigating through a document, browsing a document, viewing a document, document window, multiple document windows, checking spelling and grammar, find and replacing text, word count and thesaurus,symbols and special characters, copying and moving text, moving/copying multiple items, undo, redo and repeat, fort type, font size, font color and highlighting text, font style and effects, spacing and ligatures, lists, paragraph alignment, paragraph borders and shading, line and paragraph spacing, copying, formatting, tab stops, left and right indents, margins, page orientation and size, co;unms and column break, page breaks, new page, black page, paragraph line, section breaks, line numbers, hyphenation, page borders, page color, watermark, cover page and page number, headers and footers, clip art, screenshots, picture and graphics files, shapes, objects, positioning object, text wrapping, special effects, text box, mail merging, data source, labels, envelopes, editing tools, tracking, accepting and rejecting revisions, comments, comparing and combining documents, password protecting a document, outline view, navigating long document, bookmarks, cross-references, table of contents, picture caption, index, footnotes and endnotes, citations and bibliographies, outlines, long documents and references review.

4. **Microsoft Office Excel (Ms. Excel)**

Worksheets, data entry, numbers, text, date and time, editing data, finding and replacing data, modifying a worksheet, adding worksheets, row and columns, selecting cells, moving and coplying cells, pasting cut and copied cells, drag and drop, freeze panes, page breaks, page setup, margins, header and footer, format text, page number, current date and time, adding the name of the workbooks, manipulating worksheet data, using worksheet data, using intermediate data management tools such as filters and advanced filters, summarizing data that meets specific conditions, linking to data in other worksheets and workbooks, creating and modifying charts, creating formulas by using some of excels built-in functions, filtering and sorting excel data, pivot table and pivot charts, filtering and slicers, creating a pivot chart report, print preview, printing documents.

5. **Microsoft Power Point**

Presentation , new slides, text boxes, deleting existing slides, rearranging slides, find and replace content, spell check, content translation, language type, duplicating content, copy pasting, special characters, slides zoom in-out, setting text fonts, changing text case, changing text size, text color, text alignments, indent paragraphs, line spacing, borders and shades, formatting using slide master, saving design template, adding picture to slide, editing added picture, image adjustments, picture style, formatting added picture, screen short, shapes to slide notes, managing sections, presentation views, slide sorter view, slideshow, backgrounds, slide orientations, adding slide numbers, adding headers and footers, running slideshow, adding and formatting smartart, animations, transitions, pdf file,

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video files, image file, printing presentation, broadcasting slideshow, packaging presentation, document password, emailing slideshow.

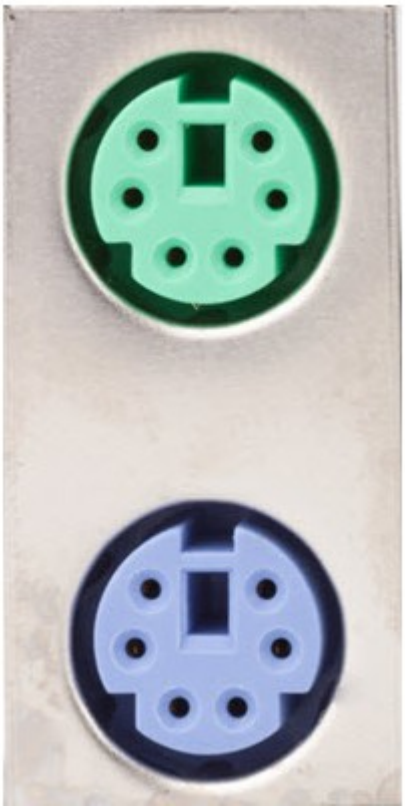
1. **Basic Hardware Connectivity and Maintenance;**

1. Computer Ports;

PS/ 2, Serial Port, Parallel Ports, Audio Ports, Video Ports, VGA Port, Digital Video Interface (DVI), Display Port, HDMI,USB, Connecting a computer to power and computer devices, Booting a computer, How BIOS work, Boot devices , Boot sequences.

1. **PS/2**

PS/2 connector is developed by IBM for connecting mouse and keyboard. It was introduced with IBM's Personal Systems/2 series of computers and hence the name PS/2 connector. PS/2 connectors are color coded as purple for keyboard and green for mouse.



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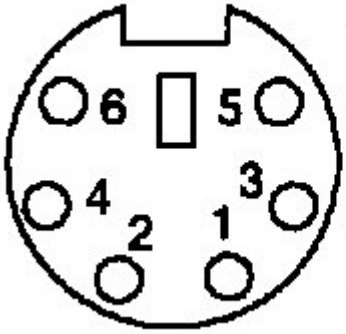
PS/2 is a 6-pin DIN connector. The pin out diagram of a PS/2 female connector is shown below.



Pinout on Female Port

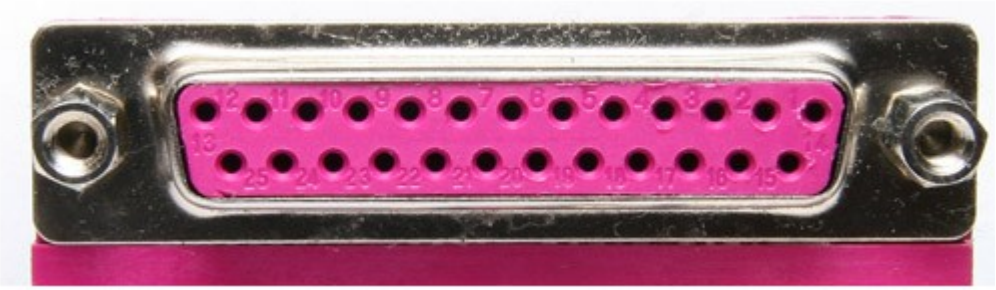
Pin Number	*Description*
1	Data
2	No Connection
3	Ground
4	+5V
5	Clock
6	No Connection

Serial Port

Even though the communication in PS/2 and USB is serial, technically, the term Serial Port is used to refer the interface that is compliant to RS-232 (Recommended Standard) standard. There are two types of serial ports that are commonly found on a computer: DB-25 and DE-9.

DB-25

DB-25 is a variant of D-sub connector and is the original port for RS-232 serial communication. They were developed as the main port for serial connections using RS-232 protocol but most of the applications did not require all the pins. Hence, DE-9 was developed for RS-232 based serial communication while DB-25 was rarely used as a serial port and often used as a parallel printer port as a replacement of the Centronics Parallel 36 pin connector.



DE-9 or RS-232 or COM Port

DE-9 is the main port for RS-232 serial communication. It is a D-sub connector with E shell and is often miscalled as DB-9. A DE-9 port is also called as a COM port and allows full duplex serial communication between the computer and it's peripheral. Some of the applications of DE-9 port are serial interface with mouse, keyboard, modem, uninterruptible power supplies (UPS) and other external RS-232 compatible devices.

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Parallel Port or Centronics 36 Pin Port

Parallel port is an interface between computer and peripheral devices like printers with parallel communication. The Centronics port is a 36 pin port that was developed as an interface for printers and scanners and hence a parallel port is also called as a Centronics port. Before the wide use of USB ports, parallel ports are very common in printers. The Centronics port was later replaced by DB-25 port with parallel interface.

Audio Ports

Audio ports are used to connect speakers or other audio output devices with the computer. The audio signals can be either analogue or digital and depending on that the port and its corresponding connector differ.

VGA Port

VGA port is found in many computers, projectors, video cards and High Definition TVs. It is a D-sub connector consisting of 15 pins in 3 rows. The connector is called as DE-15.

VGA port is the main interface between computers and older CRT monitors. Even the modern LCD and LED monitors support VGA ports but the picture quality is reduced. VGA carries analogue video signals up to a resolution of 648X480.

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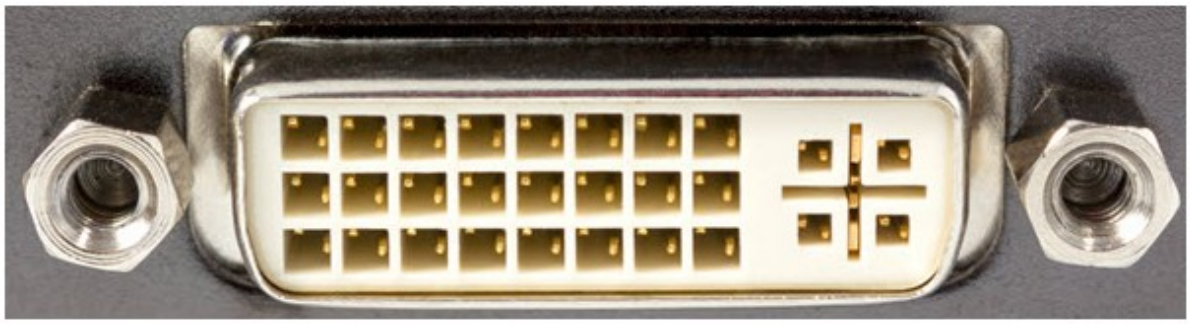
Digital Video Interface (DVI)

DVI is a high speed digital interface between a display controller like a computer and a display device like a monitor. It was developed with an aim of transmitting lossless digital video signals and replace the analogue VGA technology.



Display Port

Display Port is a digital display interface with optional multiple channel audio and other forms of data. Display Port is developed with an aim of replacing VGA and DVI ports as the main interface between a computer and monitor.

The latest version DisplayPort 1.3 can handle a resolution up to 7680 X 4320.

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Component Video

Component Video is an interface where the video signals are split into more than two channels and the quality of the video signal is better that Composite video.

Like composite video, component video transmits only video signals and two separate connectors must be used for stereo audio. Component video port can transmit both analogue and digital video signals.

The ports of the commonly found Component video use 3 connectors and are color coded as Green, Blue and Red.

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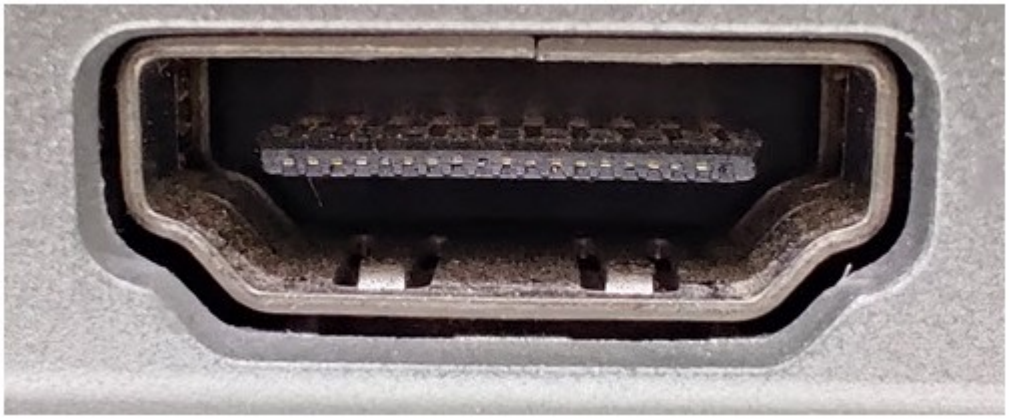
HDMI

HDMI is an abbreviation of High Definition Media Interface. HDMI is a digital interface to connect High Definition and Ultra High Definition devices like Computer monitors, HDTVs, Blu-Ray players, gaming consoles, High Definition Cameras etc.

HDMI can be used to carry uncompressed video and compressed or uncompressed audio signals. The HDMI port of type A is shown below.



RJ-45

Ethernet is a networking technology that is used to connect your computer to Internet and communicate with other computers or networking devices.

The interface that is used for computer networking and telecommunications is known as Registered Jack (RJ) and RJ – 45 port in particular is used for Ethernet over cable. RJ-45 connector is an 8 pin – 8 contact (8P – 8C) type modular connector.

The latest Ethernet technology is called Gigabit Ethernet and supports a data transfer rate of over 10Gigabits per second. The Ethernet or a LAN port with 8P – 8C type connector along with the male RJ-45 cable is shown below.

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2. Booting a computer

Introduction

Booting is a start-up sequence that starts the operating system of a computer when it is turned on. A boot sequence is the initial set of operations that the computer performs when it is switched on. Every computer has a boot sequence. The average computer doesn't understand the boot sequence but is important to know for customizing and troubleshooting your computer.

Boot Loader

Computers powered by the central processing unit can only execute code found in the systems memory. Modern operating systems and application program code and data are stored on nonvolatile memories or mass storage devices. When a computer is first powered on, it must initially rely only on the code and data stored in nonvolatile portions of the systems memory. At boot time, the operating system is not really loaded and the computer's hardware cannot peform many complex systems actions.

The program that starts the "chain reaction" which ends with the entire operating system being loaded is known as the boot loader (or bootstrap loader). The term creatively came from early designiners imagining that before a computer "runs" it must have it's "boots strapped". The boot loader's only job is to load other software for the operating system to start. Often, multiple-stage boot loaders are used, in which several small programs of increasing complexity sequentially summon one after the other, until the last of them loads the operating system.

Boot Devices

The boot device is the device from which the operating system is loaded. A modern PC BIOS (Basic Input/Output System) supports booting from various devices. These include the local hard disk drive, optical drive, floppy drive, a network interface card, and a USB device. Typically, the BIOS will allow the user to configure a boot order. If the boot order is set to:

1. CD Drive
2. Hard Disk Drive
3. Network

then the BIOS will try to boot from the CD drive first, and if that fails then it will try to boot from the hard disk drive, and if that fails then it will try to boot from the network, and if that fails then it won't boot at all.

Boot Sequence

There is a standard boot sequence that all personal computers use. First, the CPU runs an instruction in memory for the BIOS. That instruction contains a jump instruction that transfers to the BIOS start-up program. This program runs a power-on self test (POST) to check that devices the computer will rely on are functioning properly. Then, the BIOS goes through the configured boot sequence until it finds a device that is bootable. Once BIOS has found a bootable device, BIOS loads the bootsector and transfers execution to the boot sector.

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If the boot device is a hard drive, it will be a master boot record (MBR). The MBR code checks the partition table for an active partition. If one is found, the MBR code loads that partition's boot sector and executes it. The boot sector is often operating system specific, however in most operating systems its main function is to load and execute the operating system kernel, which continues startup. If there is no active partition or the active parition's boot sector is invalid, the MBR may load a secondary boot loader which will select a partition and load its boot boot secotr, which usually loads the corresponding operating system kernel.

2. Computer Applications Packages;

- Word processor
  a) Microsoft Word
  b) WordPerfect
  c) Lotus Word pro
  d) Work Pages
- Database Processing
  a) ADABAS
  b) IBM DBS2
  c) Microsoft Access
  d) Microsoft SQL Server
  e) MySQL,
  f) Oracle RDBMS
  g) QuickBase
- Spreadsheet calculation
  a) Openoffice.org calc
  b) Microsoft Office Excel
  c) GoogleDocs
  d) Gnumeric
  e) Apple iWork Numbers
  f) Lotus 1-2-3
  g) Libreoffice Calc)
- Presentation Programs
  a) Power point
  b) Openoffice.org
  c) Impress
  d) Windows Movie Maker
  e) Keynote
- Communication
  a) Internet and Email

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Desktop is your work area on which program and files icons are located. it is basically a workspace where you can access everything you need to operate your computer, such as system components, applications, the Internet, etc.

The desktop contains: **icons**

Start button **taskbar** **System Tray**

The desktop contains:

- **Start button**: one of the most important tools you will use while working with Windows XP. The Start button allows you to open menus and start applications.

- **Taskbar**: primarily used to switch between open windows and applications

- **Icons (or graphical pictures)**: represent applications, files, and other parts of the operating system. By default Windows XP provides you with one desktop icon, the Recycle Bin. Learn more about the Recycle Bin in a later module.
- **My Computer.** The My Computer icon provides access to the resources on your computer, you can access your drives and other peripherals by clicking on the My Computer icon. You can also access the Control Panel through My Computer.
  - **Internet Explorer;** The Internet Explorer icon launches the Internet Explorer browser.
- **My network places.** If you are working on a network, this displays all of the

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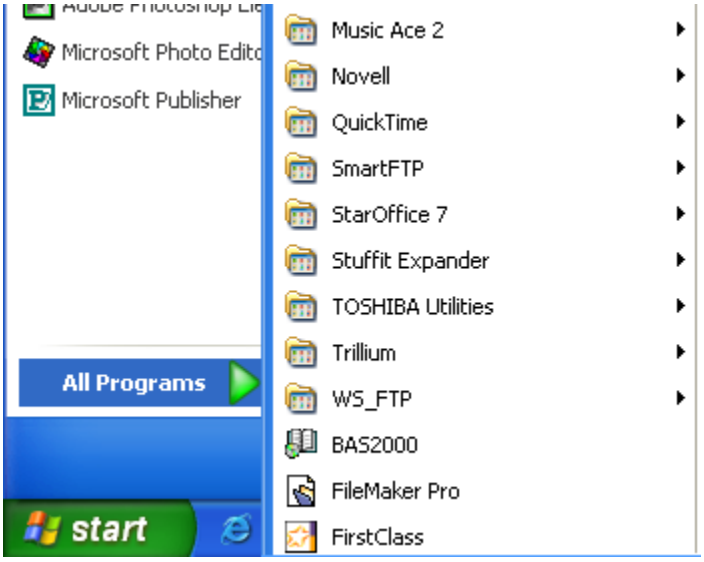
computers on the network

- **System Tray**; The part of the taskbar that holds the clock, volume control, and icons for other utilities that runs in the background of your system
- **Recycle Bin**. This is where you dump any files you want to delete.

Starting a Program

1. Click the **Start button**, the Start menu will pop up. 2. Point your cursor to **all Programs**. A Menu listing the different program categories pops up.



2. **Point your cursor** to Microsoft Office Another menu will appear to the right of the All Programs menu, as shown below. 4. Click on Microsoft Word.

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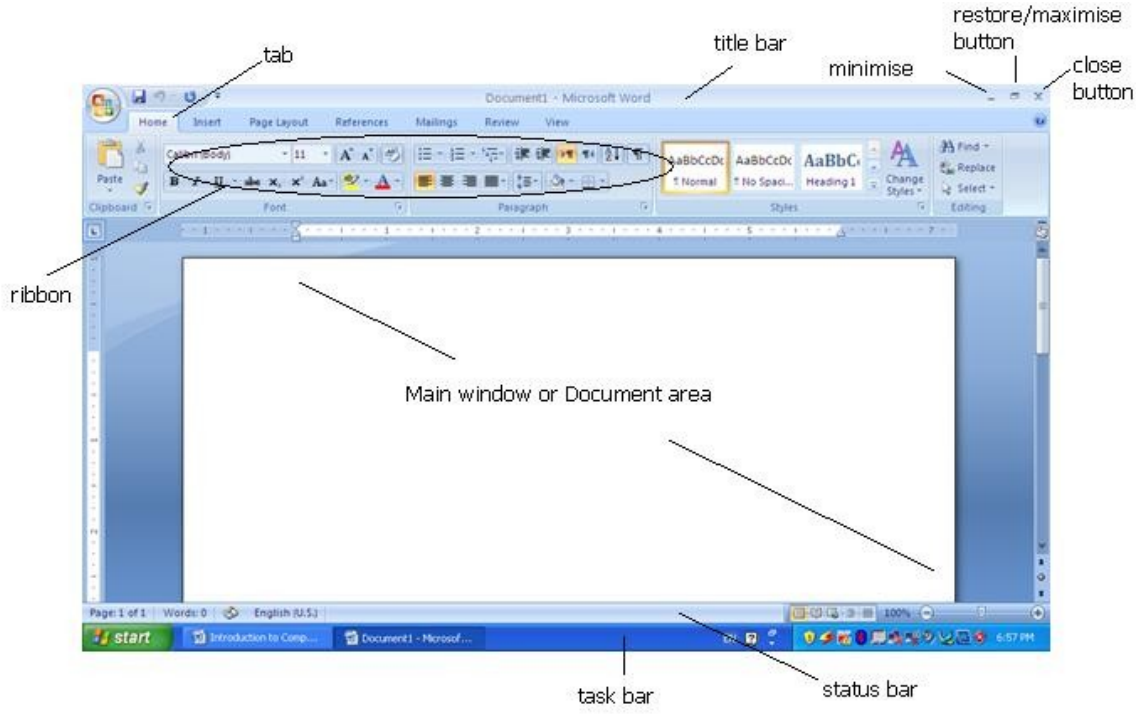
Parts of a Window

1. **Close Button** Closes the window or program, removing it from the screen and the computer‟s memory. **Minimize Button** Minimizes a program from view. The program is still ready for use and can be found in the task bar.
2. **Maximize Button** Enlarges the window so that it fills the entire screen, allowing you to see your entire workspace. Notice, when a window is maximized, this button is not shown.
3. **Restore Button** When a window is maximized, this button is shown. Clicking it will make the window smaller.
4. **Tab** Controls what a program does. The menu functions listed will change from program to program, but the bar is always located at the top.
5. **Status Bar** Displays information about the program, such as instructions or special information.
6. **Title Bar** Displays the name of the program and name of the file in use.
7. **Main Window** .This is where you work within a program. If it is Microsoft Word, this is where you would type, if it is a web browser this is where the web page would be displayed.
8. **Ribbon.** Have shortcuts to the menu items



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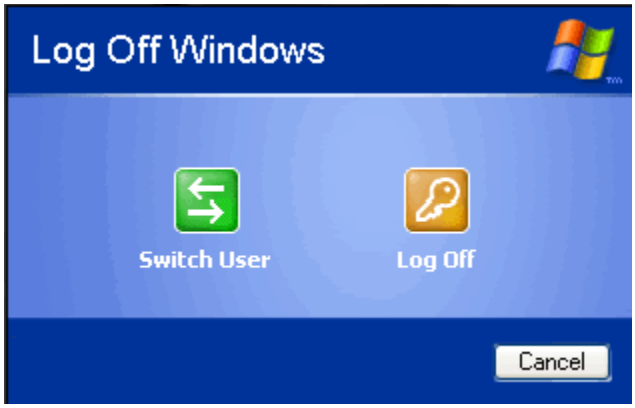
Log off and Switch Users

More than one person may use your computer. For example several coworkers may be able to access your computer on a computer network. Windows XP allows everyone who uses your computer to have separate computer accounts. A computer accounts tracks each person's unique settings, documents, and email accounts.

Windows XP even enables you to log off the computer so someone else can log on without having to restart the computer.

To log off/switch users:

Click the **Start menu** and click **Log Off**.
A dialog box appears asking you if you want to **Switch User or Log Off**. **Switch User** allows someone else to log on to the computer. If you choose to Switch User, your applications will continue to run in the background while the new user logs on.
If you choose **Log Off**, your applications will close.
In any case, you're taken to a Windows XP logon screen where you're prompted to enter your username and password.



Turn Off and Restart the Computer

When you've finished using Windows XP, be sure to turn off (or shut down) the computer correctly.

To turn off the computer:

- Click the **Start menu**.
  Click **Turn Off Computer**.
  A dialog box opens. Click **Turn Off**.

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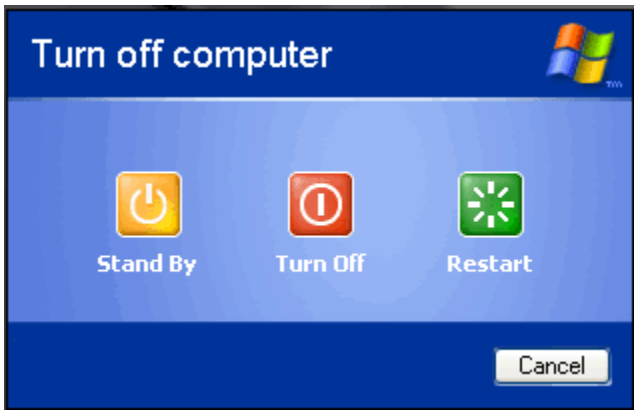

If you're experiencing computer problems or have installed something new, you can simply **restart** your computer.

To restart the computer:

Click the **Start menu**.
Click **Turn Off Computer**.
A dialog box opens. Click **Restart**.

Working With Files and Folders

Open your Folders

Double click my computer and locate your file.
Double click on the folder.
The contents of the folder will be displayed in the open window. To get back to the previous folder, click the Back Button.

Creating a Folder

- Locate the place to create your folder e.g. My Documents, Desktop, C Drive
- Go to File – New – Folder. Or if creating on the desktop right click on the desktop and use the new option to create a folder.
- The folder will appear at the bottom of your list. When it displays as a black rectangle with the words **New Folder** highlighted in blue, it is prompting you to give it a name. Type the name of your folder and hit **Enter**.

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Renaming a Folder

To rename an existing folder, move your mouse over the name and right click. A menu will appear. Close to the bottom you will see the word **Rename**. Click it, and the black rectangle with the word highlighted in blue will appear. Rename your folder and hit **Enter**.

Deleting a folder

Right click on the folder and select delete or. Select the file, go to file menu and select delete

A dialog box appears asking if you want to delete the folder, click yes

Once files are deleted they go to the recycle bin which is a holding or storage location for files not required. Such files are still in the hard disk. These files can be retrieved from recycle bin as long as recycle bin have not been emptied.

Retrieving files

Go to recycle bin.

Locate the file and right click on it and select restore.

Moving and copying files in folders

Using menu command

Select the file

Go to edit and click on copy

Select the folder

Go to edit and click on paste. By dragging

Click on the file and hold

Drag it to the folder and release the button

Procedure of creating shortcuts

Locate the item

Right click and select shortcut

Drag the shortcut created to the desktop.

To set a program to start when windows start

Right click on the start button and select open.

Click on the program or file you want and hold and drag it to the start menu on the icon for programs.

Printing

If any printers are already set up their icon appears in the printer folder (on start, then printers and faxes).

To set up a printer;

Click start menu, control panel. Select printers and other hardware.

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3. Microsoft Office Word (Ms. Word)

Creating a new document, Opening a document, Previewing, saving and closing a document

Using help, Exiting word, entering and deleting text, selecting and replacing text, Navigating through a document, browsing a document, viewing a document, document window, multiple document windows, checking spelling and grammar, find and replacing text, word count and thesaurus,symbols and special characters, copying and moving text, moving/copying multiple items, undo, redo and repeat, fort type, font size, font color and highlighting text, font style and effects, spacing and ligatures, lists, paragraph alignment, paragraph borders and shading, line and paragraph spacing, copying, formatting, tab stops, left and right indents, margins, page orientation and size, co;unms and column break, page breaks, new page, black page, paragraph line, section breaks, line numbers, hyphenation, page borders, page color, watermark, cover page and page number, headers and footers, clip art, screenshots, picture and graphics files, shapes, objects, positioning object, text wrapping, special effects, text box, mail merging, data source, labels, envelopes, editing tools, tracking, accepting and rejecting revisions, comments, comparing and combining documents, password protecting a document, outline view, navigating long document, bookmarks, cross-references, table of contents, picture caption, index, footnotes and endnotes, citations and bibliographies, outlines, long documents and references review.

Word processing and word processor

1. Word processing

   Word processing is the **activity** of manipulating text, characters, words and sentences so as to make the final document free of **errors** and also **attractive** to look at. It enables one to create, edit, revise, retrieve, save and printing of documents or store for future use

2. Word processor
   - This is a computer **application package** used in manipulation of text, characters, words and sentences so as to make the final document free of errors and also attractive to look at.
   - Also defined as a "computer application package used in processing of words."
   - We can also define it as a computer application software that enable a person to create, save, edit, format and print text documents

3. Advantages of word processors over the conventional method of document preparation
   a. Documents can be **stored** for future use
   b. Typing is easy and efficient because some actions are **automated** E.g.
   c. Has special **editing** tools e.g. spelling & grammar check, thesaurus Etc.
   d. Have superior **formatting** features e.g. underline, bold, italic, color Etc.
   e. One can use Copy, Cat and pest commands i.e. reuse text without having to retype
   f. One can easily **insert** tables and illustration e.g. pictures, clip art, charts auto shapes Etc.

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g. One can **print** as many copies of the document as he/she want
h. One can **correct** or edit text without wasting papers

4. Features of word processor

Some of the common features includes:-

a. Text editing features E.g.
    i. Copy, Cut and Pest
    ii. Find, search and replace
    iii. Undo and redo
    iv. Different Font types E.g.
        1. Size
        2. Color
        3. Bold
        4. Underline Etc.
b. Paragraph editing features E.g.
    i. Paragraph numbering and bulleting
    ii. Paragraph alignment e.g. left, center, right and justified
    iii. Paragraph and lines spacing
    iv. Word wrap, (**Format** tab, in the **Arrange** group, click **Position**)
    v. Drop cap
    vi. Paragraph Indent and Tab to align text

Paragraph and lines spacing

Word wrap, (**Format** tab, in the **Arrange** group, click **Position**)

Drop cap

Paragraph Indent and Tab to align text

a. Page editing features E.g.
    vii. Page numbering
    viii. Page color
    ix. Page setup E.g.
        1. Page margins
        2. Page orientation
        3. Page size
        4. Insertion of page columns
        5. Insertion of new page
    x. Page background E.g.
        1. Insertion of watermark
        2. Insertion of borders
    xi. Insertion of page header and page footer: - INSERT TAB (in the **Header & Footer** group click **Header** or **Footer**.)
b. Creation of tables **used** in (for)
    xii. Aligned text in columns

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xiii. Sorting of text and numbers
xiv. Perform simple calculation

c. Mail merge used in sending similar yet personalized letters to several people at the same time
d. Addition of graphs and auto shapes
e. Printing of documents
f. Protection of documents using passwords

5. Different types of word processors application exist today. They include:-
   a. Microsoft word (MS Word)
   b. WordPerfect
   c. Word Pro

   The most widely used computer package is Microsoft (MS) Word, developed by Microsoft Corporation, which is part of Microsoft Office suit.
6. Purpose of word processing (use)
   a. Writing letters
   b. Writing reports
   c. Writing projects
   d. Writing books
   e. Writing essays, memos, magazines and curriculum vitae
7. Ms Word

   It's a computer **application package** used for manipulation of text, characters, words and sentences so as to be free of errors and attractive to look at
8. Starting MS Word

   One can start MS Word in the following three ways
   a. Click **start button** → then select **All Programs** → then choose **Microsoft Office** → then select on the **Microsoft word 2007**
   b. Double click on the **MS word icon** in the Desktop if it's available and it will open word
   c. Open any **word Document** and it will start MS Word automatically
9. Elements found in Ms Word

   i. **Title Bar** – located at the top of the window, Display name of the application and the name of the document
   ii. **Tab Bar** – located below title bar, list available options in Ms Word
   iii. **Tool Bar** – located below the Tab bar, provide access to commands used in Ms Word (for word processing task)
   iv. **Formatting toolbar Ruler** – located below Tool Bar, it is used to set tab stops and indent
   v. **Scroll Bar and scroll buttons** – used to view different areas of the active document, up and downwards or side by side
   vi. **Status Bar** – located across the bottom of the window, display information about the active document e.g. current page number, number of words Etc.
   vii. **View buttons** – located directly to the right side of the status bar, provide a quick way of moving between various page view

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- **HOME TAB** contains command buttons that you can use to carry out many of the commonly used word commands
- You can access all Ms Word **commands** through the **Tab** Bar
- When you want to **save a document**, click **Office button**, → then **save** or **save As** (or press **CRT + S** at the same time)
- In the **SAVE IN** dialog box, select where you want to save the document
- When naming a document, don't use the following characters / : ; *, ?, <, >

10. Home Tab

Contain the most commonly used tools for word processing tasks

11. Insert Tab

Used to insert most of the items to our document e.g. blank page, tables, illustration, links, headers and footers, symbols Etc.

12. Page layout Tab

Use to change the layout of the page, e.g. page setup, page background, paragraph layout etc.

13. Opening a document on the hard disk

Start word normally then:-

- Click **Office Button**,→ then select **Open** to open a dialog box
- In the **look in**, select where the document you want to open is located and then
- **Select** the document and click **open button** to open it
- **Keyboard shortcut** To display the **Open** dialog box, press **CTRL + O**.

Or

Click the Start button, → recent document or

Office button, → then **open** dialog box, → then **recent documents**

14. Creating a new document

Click office **Button**, → then select **New Document**, → then **Blank document**, → then click **Create**

15. Ways of inserting text

Word is a **WYSIWYG** (what you see is what you get) word processor. This mean that what you see on the screen is exactly what will be printed

16. Using the ENTER key

It is used to:-

i. End a paragraph
ii. End a short line E.g. a title, heading or a list of items
iii. Create a blank line

17. Using the Tab key

Use to align text to specific column positions. Space bar creates a jagged appearance when printed.

18. Using the Backspace and Delete Keys

- Backspace key is used to delete characters to the left of the cursor key
- Delete key is used to delete characters to the right of the cursor key

19. Saving a new document

a. Click **office button**, → then **save As**
b. Type the new document name in the File Name (don't use / , : ; * ? < >)

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c. Select the location from the **Drop Down** list of locations in the **Save In** box
d. Click save

To save existing document, simply click on the save icon or press **CTRL + S**

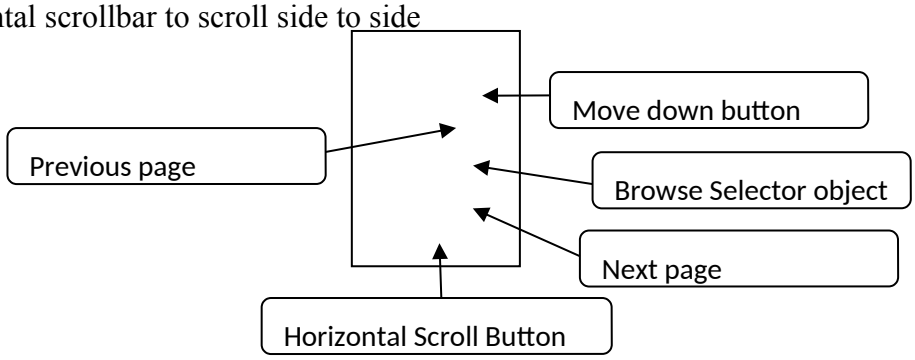
To save all open documents at the same time, hold down SHIFT key and click save all on the office button

20. Closing a document

- Click office button, → then close
- Press shift key, → then close all
- Click close button on the upper right corner of the Ms Word window

21. Scrolling in a document

- ✓ Use vertical scrollbar to move the view up and down
- ✓ Use horizontal scrollbar to scroll side to side



22. Moving the **insertion point** in a document

- One can use the mouse and click anywhere you want or
- **Use the key board as follows**

a. Page Down → Down one screen
b. Page Up → Up one screen
c. Ctrl + End → To the end of the document
d. Ctrl + Home → To the top of the document
e. End → To the End of the line
f. Home → To the beginning of the line

23. Inserting text in insert mode and overwrite mode

- By default, word operates in insert mode. New text does not replace existing text but bushes it to the right as you type.
- When you need to overwrite text, change to overwrite mode by pressing the **Insert** Key (note that the status bar OVR is now active)

24. Select text with a **mouse**

a. **Any amount of text** – Hold down the mouse button and drug over the text
b. **A word** – double click the word
c. **A line of text** – move the mouse pointer to the left margin of the line until it changes to a right-pointing arrow, and then click
d. **Multiple line of text** – move the mouse pointer to the left margin of the line until it changes to a right-pointing arrow, and then drag up or down

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e. **A sentence** – Hold down the **CTRL** Key and click anywhere in the sentence
f. **A paragraph** – triple-click anywhere in the paragraph or – move the mouse pointer to the left margin of the paragraph until it changes to a right-pointing arrow, and double-click the mouse.
g. **Multiple paragraphs** – double click and while holding down the Shift Key, drag up or down **/or/** move the mouse pointer to the left margin of the paragraph until it changes to a right-pointing arrow, and double-click the mouse. then while holding down the shift key drag up or down
h. **A large block of text** – click the start, scroll to the end of the selection and while holding down **SHIFT** Key, click at the end
i. **An entire document** – press **CTRL + A**, at the same time

25. **Deleting text**

    Use BACKSPACE key to delete characters to the last or use DELETE key to delete characters to the right of the insertion pointer.

26. **Replacing text**

    One can type the new text to replace the old text

Text editing features

1. Copying and moving(cut) text
   - We use **CUT** to <u>move</u> text and **COPY** to make another copy of text
   - We can either use the mouse by **right clicking** on the **selected** text or we can use the **Tool Bar** in the **Home Tab** on the **clipboard group**
2. Text formatting

   In the **Home** Tab, **Font** section, one can format font:-

   i. Size
   ii. Style
   iii. Color
   iv. Underline
   v. Super script and Subscripts ETC.
3. Find and replace

   One can find text in the document by using the **FIND** feature in the tool bar in edit section of the Home Tab
4. Spell check and thesaurus
   - **Thesaurus** is used to find synonyms, antonyms or related words
   - Spelling errors are underlined in red while grammatical errors are underline in green
   - Ms Word checks the following errors

     i. Spelling errors
     ii. Repeated words
     iii. Irregular capitalization
     iv. Errors in sentence structure
     v.

Paragraph formatting

1. Paragraph numbering and bulleting

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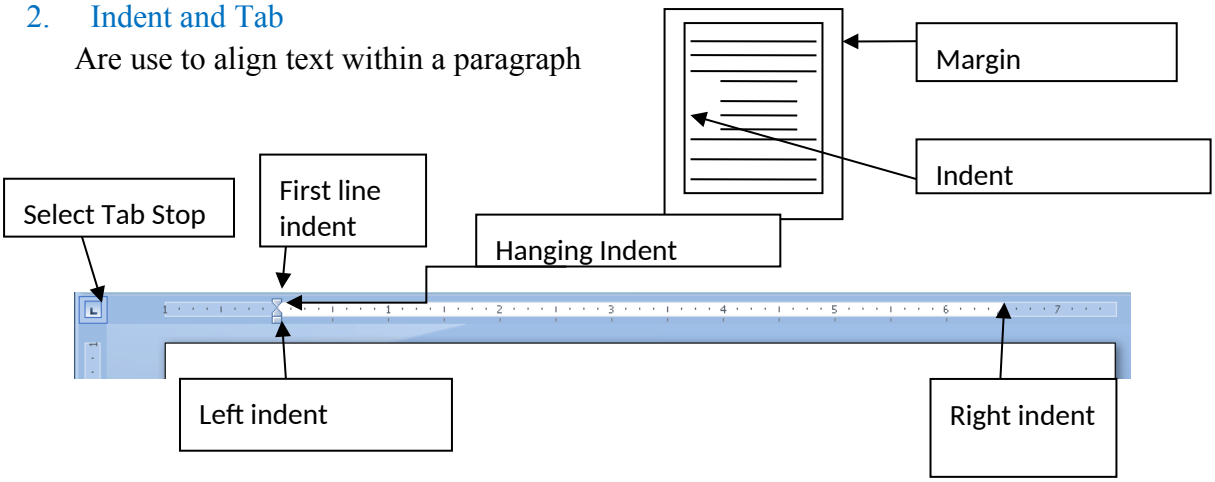
- Select text you want to add bullet/numbering and click bulleting/numbering
- One can increase or reduce indent of numbering

2. Indent and Tab

Are use to align text within a paragraph



3. Word wrap

Word Wrap enable one to easily place text around pictures, shapes and tables with any position or style that you want e.g.

a. Square
b. Tight
c. Behind or in front of text Etc.

4. Drop cap

Use to create a large dropped initial capital latter. It is sometimes used when starting a paragraph at the beginning of a chapter in novels

To create, click the **paragraph** you want to begin with a **drop cap**, on **the Insert tab**, → click **Drop Cap** in the text section for the dialog box to appear. Then select the one you want

Page formatting

1. Page margins
   - Page margins are used to restrict the printable area of a page
   - When changing the page margin, we follow the following steps:-
     i. Switch to **page layout** tab, → select **margin**s on **page setup** section and select the option available /or/
     ii. Point to a margin boundary on the horizontal ruler or vertical ruler to change each one of them
     iii. When the pointer changes to a margin height pointer, drag the margin boundary
   - To specify exact margin measurements, select the "**page layout**" tab and click the **margins,** type the required values for each of the margin needed to be changed
2. Insert page breaks

   Click page **Layout Tab**, → select **BREAKS** in the page setup section
3. Setting column layout

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- Columns are used to format document into layout, like the columns of a newspaper
- To quickly format text into columns, select the text to be included in the columns, click on the columns icon in the page **layout tab**, → page setup section, and highlight the number of columns you want

4. Adding header and footer

Header and footer are typically used in printed documents. They appears above the top margin or below the bottom of each page margin respectively

One can include graphics or text e.g. page number, date, company logo, document title or authors name etc.

To insert page header and/or footer, follow the following steps

i. Click **Insert Tab**, → go to **header & footer** section, the select header or/& footer then
ii. Select the format of page header/footer you want
iii. Enter text, graphics Etc., in the header/footer

5. Page orientation

Page orientation refers to the direction the page text is laid out on the page Eg. Portrait or landscape

i. Portrait – text is laid across the upright page
ii. Landscape – across the horizontal page

Steps are

i. Click the Page **Layout Tab**, → select page **setup section**, → then select orientation

Tables

1. Tables

It is made up of **ROWS** and **COLUMNS** of cells that you can fill with text and graphics

You can use tables to:-

i. Align text in columns
ii. Sort text
iii. Perform simple calculation

2. Create a table

- Select **Insert Tab,** → click table in table section
- Select the number of rows and columns of the table and then click okay

3. Edit table

One can perform the following editing features on a table namely:-

i. **Resizing** Row/Column size – using mouse to drag the boundaries of the Row/Column /or/ use the **Layout Tab,** → **Cell Size** Section
ii. **Insert** Row/Column – using the mouse, select Row/Column that you want to add a Row(above or below) or Column (to the right or left), then **right** click on it and then select add Row/ Column /or/ use the **Layout Tab,** → **Rows & Columns** Section

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iii. **Delete** cells/Rows/Columns – using the mouse, select Row/Column that you want to delete, right click on it and then select delete Row/Column /or/ use the **Layout Tab, → Rows & Columns** Section

iv. **Merge** Cells/Rows/Columns – using the mouse, select Cells/Row/Column that you want to merge, right click on it and then select merge Row/Column or use /or/ use the **Layout Tab, → Merge** Section → merge

v. **Split** Cells/Rows/Columns – using the mouse, select Cells/Row/Column that you want to merge, right click on it and then select merge Row/Column or use /or/ use the **Layout Tab, → Merge** Section → split

4. Formatting table
   - You can use all the text formatting options to format TEXT within a table
   - To give a table more polish look, you can use
     i. **Borders** – table **Design Tab**, → then **Table Styles**
     ii. **Shading** – table **Design Tab**, → then **Table Styles**
5. Performing arithmetic calculation on tables

   One can perform simple arithmetic on data entered in a table by following the steps below

   i. Click the cell in which you want the results to appear
   ii. On the **Layout Tab**, → **Data** section, select **formula**
   iii. In the **paste function** box, click a function e.g. sum
   iv. In the **Number Format** box, enter a format for number, E.g. Decimal or Percentage
   v. In the **Formula box**, type the cell references in the parentheses and then click **OK** button E.g. =Sum(Left)

Other features

1. Sorting

   Used to sort text, numbers or dates within a table or in the list either in ascending or descending order
2. Mail merge
   - Used when you want to send out a standard yet personalized latter to several people at the same time
   - Instead of repeating the latter with different address, you can create a form containing the address list and then merge it with either Envelope, labels, latter, E mails, Directories Etc
   - When creating the Mail Merge, follow the following steps:-
     i. Create the main document E.g. Envelop or a Latter
     ii. Create the Data Source → **Mailings Tab**, → **Start Mail Merge** section, **Select Recipients**

     One can either
     - Type new list
     - Use an existing list

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- Or select from Outlook content

iii. Insert mail **Merge Field**

iv. Previewing & fine tuning the merge Document

3. Inserting a picture objects

- Word Processing documents may have pictures objects that can be Drawing (Clip Art), Charts (graphs), Photographs, Etc.
- Ms Word comes with a Clip Art Gallery that includes a wide variety of Clip Art that make it easy for you to dress up your document with professional designed images E.g. people, nature, human activity, maps, photographs etc. one can :-

i. **Insert a picture** from another file. → click **Insert** Tab, → select **Picture** in **Illustration** section

ii. **Insert a Clip Art** file. → click **Insert** Tab, → select **Clip Art** in **Illustration** section. In the Clip Art Dialog box, select the one you want

iii. **Draw objects** → click **Insert** Tab, → select **Shapes** in **Illustration** section and select any object to draw

4. Word art

This is used to add special effects to the text

With Word Art, one can

- Create shadowed text
- Create skewed text
- Create Rotated Text
- Create Stretched text
- Create text fitted to predefined shapes

Changing normal text to Word Art

- Select the text to change into **Word Art**, usually a short phrase. Then click the **Word Art** icon in the **Insert** Tab, → **Text** section

5. Section Breaks

A Section Break is a mark you insert to show the End of a section, E.g. book chapter. The section includes such information as margins, page orientation, header, footer and sequence of page numbering.

There are three types of **section breaks** in Ms Word

- Next Page – start new section on new page
- Continuous – start new section on the same page
- Odd or Even Pages – start new section on the next Odd or Even page
- Column break

To start Section Break, go to **Page Layout** Tab, → **Breaks**, in the **Page Setup** section

6. Table of contents

Table of Content (TOC) provides an overview of the topics discussed in a document and is used to quickly navigate to a give topic in a document. It uses the Heading Style to determine deferent levels.

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To create TOC, you 1st ensure that you apply similar heading styles (Header 1 through Heading 9) to the heading you want to include in the Table of Content and the :-

i. Move to the top of the document
ii. Click **References** Tab, → select **Table of Content**
iii. Scroll down the list of available styles

7. Printing a document/s

Printing is the process of making a hard copy of the document on a real paper. Before printing, the following factors should be considered:-

i. Previewing the document to display each page as it would appear when printed
ii. Before printing, one should know the page orientation of the document
    a. Portrait – across paper in the upright position
    b. Landscape – across the paper in horizontal position
iii. Print options E.g.
    a. Print a **range** of pages
    b. Print **multiple** pages
    c. Print only **ODD** or **EVEN** pages
    d. Print the document on different paper **SIZE**

Print several documents **together**.

4. Microsoft Office Excel (Ms. Excel)

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Worksheets, data entry, numbers, text, date and time, editing data, finding and replacing data, modifying a worksheet, adding worksheets, row and columns, selecting cells, moving and coplying cells, pasting cut and copied cells, drag and drop, freeze panes, page breaks, page setup, margins, header and footer, format text, page number, current date and time, adding the name of the workbooks, manipulating worksheet data, using worksheet data, using intermediate data management tools such as filters and advanced filters, summarizing data that meets specific conditions, linking to data in other worksheets and workbooks, creating and modifying charts, creating formulas by using some of excels built-in functions, filtering and sorting excel data, pivot table and pivot charts, filtering and slicers, creating a pivot chart report, print preview, printing documents.

Spreadsheet

1) Definition
   i. Spreadsheet is a computer **application package** used for **manipulating numerical information**
   ii. It consist of a series of Rows and Column
   iii. Intersection of a row and a column is called a **CELL**
   iv. Figures and text can be inserted into the cell
   v. Every cell is **referenced** by the column **Label** and Row **Number**
2) Manual verses electronic spreadsheet

MANUAL	ELECTRONIC
i. Manual calculation	i. Electronic/automatic calculation
ii. Difficult to correct or edit figures	ii. One can easily play with figures
iii. Hard to insert charts, pictures and graphics	iii. Easy to insert charts, pictures and graphs

3) Facilities provided by electronic spreadsheet
   i. **Formulas** – used to calculate new values automatically E.g. sum, product, average Etc.
   ii. **Data formatting features** – both text and numerical data can be formatted using pre-defined format E.g. fonts, size, styles, decimals, currency, date Etc.
   iii. The ability to adjust column widths and row heights, hide rows/column and freeze panes
   iv. **Automatic recalculation** – when you change one figure, the rest of the figures are automatically recalculated by the computer to correspond with the different inputs
   v. Data **sorting** and **filtering** capabilities
   vi. **Chart** making E.g. line, bar, charts, pie charts or histograms
   vii. Data **validation**
   viii. **Printing** of entire or portion of a document
4) Components of a spreadsheet
   They include
   i. **Work sheet** – cells organized into Rows and Column. Groups of worksheets is called **Workbook**
   ii. **Data base** – list of items stored in a worksheet
   iii. **A graph/chart** – can be created from data stored in a worksheet
5) Application/uses of spreadsheet
   i. Used for storing data records

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ii. Used for numerical calculation
iii. Researchers can compile and analyze their results using spreadsheet
iv. Teachers can compile their student's marks and produce overall results
v. Data management. E.g. Sorting, Filtering and tallying
vi. "What if" feature can be used to forecast and outcome.

6) Microsoft Excel

Microsoft Excel is one of many type of spreadsheet Application Program for manipulating numerical information

7) Starting excel

One can start MS Excel in the following 3 ways

a. Click **start button** → then select **All Programs** → then choose **Microsoft Office** → then select on the **Microsoft Excel 2007**
b. Double click on the **MS Excel icon** in the Desktop if it's available and it will open word
c. Open any **Excel Document** and it will start MS Excel automatically

8) Exiting excel

There are 2 ways

i. Click **office button**, → **Exit/close** button
ii. Click application's **close** button

9) Elements of Excel Window

i. **Title** Bar – contains the name of the workbook
ii. **Tab** Bar – shows available options for working with the spreadsheet
iii. **Formula** Bar – contains the name of the current cell/range and the date you have type in it
iv. **Column labels** – letters across the top of the worksheet which identify different columns in the worksheet
v. **Row numbers** – numbers down the side of the worksheet used to identify the Rows in a worksheet
vi. **Worksheet** Browsers – for navigating across/through worksheet.
vii. **Status** Bar – indicates current activity e.g. Editing, Enter, Reads, Saving etc

10) Features of Excel Window

i. **Worksheet** – one sheet in a workbook. It's made up of Rows and Columns of information
ii. **Workbook** – this are user file that **hold** information. It consist of several worksheet
iii. **Column** – **vertical** divisions of a worksheet that are identified by letters (A, B, C, ….. Z, AA, AB, …..)
iv. **Row** – **horizontal** divisions of a worksheet that are identified by numbers. They are 65536 Rows in a worksheet
v. **Cell** – **intersection** of a Row and a Column. It has an address that consist of a column letter and a row number
    - The current (Active) cell is highlighted by a thick borders and its address is displayed in the address bar
vi. **Label** – **text** entry into the worksheet that consist of alphanumeric characters that provide descriptive information (E.g. Names) and have **no** numerical/mathematical significance in Excel
vii. **Value** – data that **has** numerical/mathematical significance E.g. Numbers, Date, Time

11) Creating a worksheet

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- When you start Excel, the program opens with a blank workbook titled "Book 1" by default. The new workbook contains 3 worksheets by default
- You can also use **Office Button** to create a new workbook as follows:-
  - Office Button, → New, → Blank, then click create

12) Opening an existing worksheet

- Click **Office Button**, → **Open**, then brows to the location where the file is located, then select Open
- Double click any Excel document and it will open Excel automatically

13) Entering Data

i. Numbers – contains characters like 0 → 9, +, -, (), /, &, %, E etc. **NOTE:-**
   - To enter a fraction, type the whole/1st number (even if it is a zero), then a space and the fraction (numerator, back slash sign "/", and a denominator) - If you omit "zero" the fraction will be interpreted as a short date.
   - The cell filled with the # symbol indicates that the column is not wide enough to display the number.

ii. Text – can include a combination of alphabets, numbers & symbols up to a maximum of 255 characters in a cell. To type a **number as text** ,precede the with an apostrophe (')

iii. Column and Rows labels – are used to **identify** your data e.g. Name

iv. Date and Time - date and time are converted by Excel into a serial number

v. Entering Data in a selected Range – **select** the range in which you want to enter data, the 1st cell in the range is active and appears with a white background. Type data in the 1st cell and press **enter** to move down one cell or **Shift + enter** to move up one cell or Tab to move right or **Shift + tab** to move left
   - Excel can also perform
     i. Autocorrect
     ii. Entering **Data series** for entering **sequences** of values automatically. You can **auto fill** date, year, positive or negative range. To start, enter number in the 1st two calls of the range and drag the cross pointer to the end of the range you want.

14) Saving a Workbook

- Click **Office button**, → then **save** or **save As** (or press **CRT + S** at the same time)
- In the **SAVE IN** dialog box, select where you want to save the document

❖ When naming a document, don't use the following characters / : ; * ? < > " | \

- you can also click **Save Button**

15) Closing a Workbook

i. Click office button, → then close/Exit

ii. Press shift key, → then close all

iii. Click close button on the upper right corner of the Ms Word window

16) Performing simple Calculation

- **Formulas** are used to perform calculation e.g. Addition, subtraction, multiplication and division
- Formulas include cell addresses that reference to the cell on which you want to perform the calculation and the mathematical operation
- The more complex calculations are left to Excel **Function**
- The formula begins with the Equals sign (=) which tell Excel that it's a calculation operation

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- The natural order of performing mathematical operation are:-
  i. Whatever is in the **Brackets**
  ii. Calculate the **Exponential** (^)
  iii. Calculate **Multiplication(*)** and **Division(/)**
  iv. Calculate **Addition**(+) and **Subtraction**(-)
- In case of **Multiplication/division, Addition/Subtraction,** the operations will be executed in the order they appear in the formula from left to right
- Operation are:-
  i. Exponentiation
  ii. Multiplication
  iii. Division
  iv. Addition
  v. Subtraction

17) Entering Formulas
  i. **Select** the cell where you will place the formula
  ii. Type the **Equal** sign (=)
  iii. **Enter** the appropriate values (), when finish, press Enter

18) Using the Auto Calculate on the status Bar

Select the cells and looking at the Status Bar, to select different formulas, **right-click** the **Auto-calculate** area of the Status Bar

19) Displaying Formulas
- Formulas are not normally displayed in the cell; instead results of the calculation are displayed. To view the formula, select the cell and look at the Formula Bar
- One can also go to **View Tab, →formula**

To edit the formula, use the **Formula Bar**

20) Excel Error Message
- Error occur when a formula cannot resolve to a **Value answer**
- Some of the errors include
  i. #DIV/0 – The formula is attempting to divide by zero
  ii. #N/A – refers to a cell that contain no value or #N/A entry (null)
  iii. #NAME? – does not recognize a Function, Range or Name
  iv. #NULL! – the formula refers to an invalid Range (remember to use commas and not space btw functions argument)
  v. #NUM! – problem with the number used in the formula
  vi. #REF! – cell reference in the formula is incorrect
  vii. #VALUE! – wrong type of argument/operation e.g. wrong syntax of the formula
  viii. CIRCULAR – appears on status bar, the formula refers to the same cell in which the formula is written

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21) Cell Reference

Cell referencing is the **location identity** of a cell or a range of cells on the worksheet. They include

Type	Refers to
B6	Cell column "B", row "6"
A10:A20	Range of cells in column "A" from row "10" through "20"
B15:E15	Range of cell in Row 15 from columns "B" through "E"
5:5	All cells in row 5
5:10	All cells in row 5 through 10
H:H	All cells in column "H"
H:J	All cells in column "H" through "J"
Sales!(C20:C30)	Information stored in sheet "Sales" from range C20 – C 30

22) Naming Cell, Formula, Constant or Range

i. Select cell, range or non-adjacent cells that you want to name
ii. Click the name box in the formula bar
iii. Type the name of the cell e.g. "Rongai" and press Enter /**OR**/
iv. Right click the selected range and select "**NAME RANGE**"

23) Relative Cell Reference

- Formula are based on their position relative to the Cell/Range that contains the formula
- When you copy a formula that uses Relative cell reference, the pasted formula will refer to a different cell, but with the same relative position with the original formula

24) Absolute Cell Reference

- The cell referred to is the same, no matter which cell refers to it
- To make a cell reference absolute, type the dollar sign"$" before the column latter and before the Row number e.g. =$A$2 i.e. B2=$A$2*10

25) Mixed Cell Reference

- A reference that is only partially absolute e.g. A$2 or $A2 is referred to as mixed cell reference
- When used/copied to another cell, only the relative part of the reference is adjusted

26) Using Functions

- A function is a ready-made formula, built into the spreadsheet to perform a series of operations on a specified value/range of value e.g. =AVERAGE (A1:A10)
- There are many functions, categorized into:-

Financial	Mathematical	Trigonometrical
Date	Time	Statistical
lookup	Reference	Text
Logical	Database	Information Function

- A **function** consist of three element
  i. The **"="** sign – indicates the start of a function
  ii. **Name** – e.g. SUM → indicates the operation to be performed
  iii. The **Arguments** – values to be acted upon by the function and are enclosed in brackets
- Multiple arguments are separated by commas
- Some functions e.g. "=NOW()", have no arguments while others have it e.g. =YEAR(37245) or =AVERAGE(F2, F3, G1:G4, H5:H8)
- commonly used excel function are:-
  i. AVERAGE – calculates the mean/average e.g. =AVERAGE(A1:A10)

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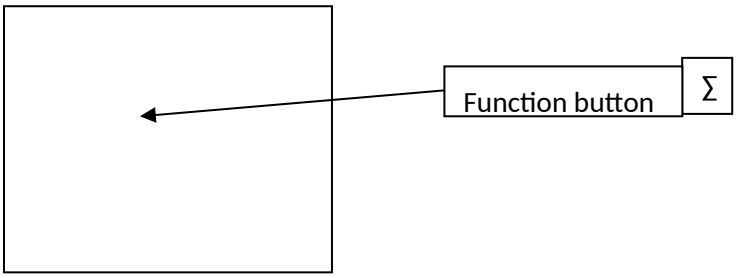
ii. COUNT – count the number of cells that contain values in the selected range e.g. =COUNT(B1:B5)
iii. MAX – returns the maximum value in a range of cells e.g. =MAX(B4:B2O)
iv. MIN - =MIN(D1:D6) – Returns the maximum value
v. SUM - =SUM(A1:B5)
vi. IF - =IF(A5>1500, "Bonus", "No Bonus") – place a conditional function in a cell
vii. PMT – =PMT(0.125/12, 120, 150,000) – Calculates the monthly repayment over time e.g. (120 monthly repayment)at 12.5%(0.125) interest par year (0.125/12) for Kshs 150,000/=

27) Using the Insert Function Feature\

i. Click the cell where you want to place the function
ii. In the "Home" Tab, → "Edit" section, click "Insert Function button"



iii. Select the function that you want to use and click "Ok"
iv. Enter the range of cells to use as Arguments in the function or click "Collapse button" to take you to the worksheet
v. Click the "Expand" button to go back to the function Arguments
vi. As you fill in arguments, partial functions results are displayed and then click okey

28) Worksheet Formatting

- Used to **Enhance** the appearance and improve readability of the worksheet
- You can format a worksheet either before or after you enter the data
- One can format text e.g.
  i. Aligning text Horizontally or vertically
  ii. Centering across multiple cells
  iii. Justifying line of text
  iv. Rotating text
  v. Shrinking text to fit in a cell – (shrink to fit) alignment options
  vi. Wrapping text in a cell
- One can change font attributes E.g. BOLD, ITALIC, UNDERLINE etc.

29) Formatting Numerical Data

One can use different number formatting/style options e.g.

Currency E.g. $400	Increase decimals	
Percentage E.g. 48%	Decrease decimals	
Commas style E.g. 1,245,766	Text	
Fraction E.g. 5/7	Scientific 5.823E+3	

30) Border and Lines

- Add borders around cell to add emphasis, define data entry area or mark Total and Sub-total

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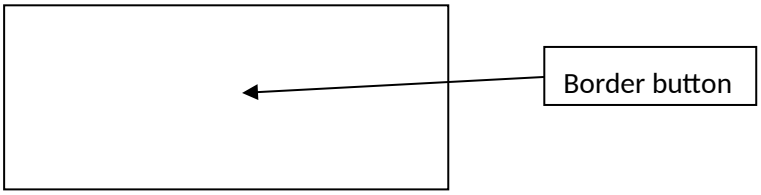
- When combined with shading, borders make your worksheets easier to read and interesting
- To add borders, follow the following steps
  i. Select cell or range to which you want to add a line or border
  ii. Click the drop-down arrow of the "Borders button" on the Home tab, → Font section and click on the desired border style



31) Colors and Patterns

Colors and patterns can help clarify meaning and emphasize important data

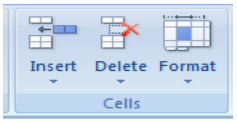
32) Adjusting Column Width

- One can use the drag feature of the mouse
- You can auto-adjust a column to fit the widest text by Double-clicking the right boundary of the column
- To size multiple columns at the same time, select the columns and drag the right boundary of one of the selected columns /or/
- To size all columns, select all button (the gray rectangle just to the left of the column letter), then drag the right boundary of any column. The **same apply** to Rows

33) Inserting and Deleting Cell, Range, Columns, Rows and Columns

Follow the steps below

i. Highlight the cell/range/column or row you want to insert/delete
ii. In the Home Tab, Click "**insert**" cell/row/column(s) to insert or Delete to "**delete**" a cell/range/row or column(s) in the **cells** section



iii. Specify the direction and click okay

34) Charts

There are many types of charts namely:-

i. Pie chart – used to show the relationship among parts of a whole so that one can compare items readily
ii. Column chart – illustrate individual values at a point in time and is suitable for side-by side comparison of numerical values
iii. Bar chart – similar to column chart but display data horizontally rather than vertically
iv. Line chart – display linear and unbroken change in large values. use to illustrate trends and changes of values over time
v. XY(Scatter) chart – similar to line chart, use to show heavy occurrence or absence of data
vi. Area chart – show change in values over time. Use to give a bigger sense of the **size** of a change or concept

- One can

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i. Create and save a chart – select the data that you want to include in the chart including the column and/or row labels that you want to include in the chart
ii. Select a chart objects – E.g. title, legend, data series etc.
iii. Add new chart data – point to the edge of the selected range until the pointer changes to an arrow, the drag the data anywhere onto the chart and release the mouse button
iv. Change the chart style – click on the chart and choose the new chart type you want
v. Add a title and legend
vi. Format chart title and legend
vii. Format the chart Axes
viii. Format chart data series

35) Sorting and Filtering Data
i. Sorting – it's **arranging** data according to a specified **field**, in ascending or descending order
ii. Filtering data – it's displaying data that **only** meet a given **criteria**
It's when you have a large worksheet and you are only interested in a small portion

36) Subtotal in a List
i. Sort the list by the column that you want to calculate sub-total
ii. Click a cell in the list
iii. Select Data/sub-totals – dialog box appears
iv. Click ok to return to the worksheet
v. Populate the sub-total dialog box

37) Printing worksheets
Preview the document so as to see how it will look like when printed
Change the paper setup to match the paper in which it will be printed on E.g.
i. Page size
ii. Page orientation
iii. Page margins etc.

Database Processing

1. Database
Database is an organized collection of data. It's considered organized because data is stored in categories that are accessible in a logical/particular manner
2. Components of database
i. **Files** – a file is a large organized data structure that contains related data E.g. Records. A file stores data and programs in the secondary memory
ii. **Records** – it's any entry of data in a database. It can also be defined as a "group of related data about on item"
iii. **Field** – a column of similar information /or/ a group of vertically arrange data that contains similar information of the same data type
3. Database system
Database system is comprised of 4 major components namely:-
i. Data – values physically stored in the database
ii. Database Hardware – this are computer hardware on which the database resides E.g. Secondary storage disk, together with associated devices e.g. CPU, channels ect.

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iii. Database Software – it contains DBMS which is an application program that assist users to store information in an orderly manner for timely and quick retrieval. It provide all the functions needed to define the Data Structure and the way they are store

iv. Database user – database users are classified into three broad categories namely:-

a. Application program
b. End user
c. Database Administrator

4. Type of databases

Category	Database package
Desktop Database	Ms-Access, Dbase, FileMaker Pro, Clipper, FoxPro, Lotus Approach
Server Database	SQL server, My SQL, Oracle, Ingress, DB2,

5. Entity-relationship modeling

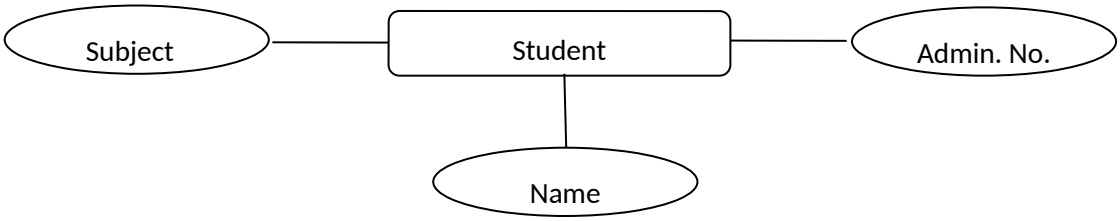
a. Entity

- An entity is a thing or an object of significant, either real or imaginary, about which information need to be known and stored
- An entity maybe an object with physical existence E.g. person, house etc. or a conceptual existence E.g. Job, Company, Country etc.
- An entity has partial properties/characteristics/**attributes** which describe it

b. Attribute

- An attribute is any information that serve to **Qualify**, **Quantify**, **Identify**, **Classify**, or **Express** the **state** of an **Entity**
- It can also be defined as "Any description of a thing of significance E.g. Text, Number, Picture, Feelings, Smell, and Sound etc."
- **Primary Key** is an Attribute that is used to uniquely identify an entity/record(identify a row in a table) and is chosen from **candidates Key**

Student Entity with attributes(Name, Address, Admin. No.)



6. Relationship among data

- Relationship is "An association among entities". It may also associate an entity with itself
- There are three types of relation namely:-
  i. **One – to – one** relationship e.g. student and admin. No.
  ii. **One – to – many** relationship e.g. student and subject
  iii. **Many – to – many** relationship

7. Database model

- Database model is a "Specific method for describing the structure and processing within a database"
- The three most used models are:-
  i. **Hierarchical** model – establishes a parent-child relationship. The **origin** of data is called the **Root**

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ii. **Relational** model – establishes relationship among entities. Data is represented in a simple **Row-Column (**tabular**)**format

iii. **Network** model

8. Features of relationship
   i. Each Row/Record in a relation contain data about the same Object/Entity
   ii. Each Column/Field contains data about an attribute/property of an Object/Entity
   iii. Each Column/Field in a table has a unique Name/Label
   iv. The order of the Field/Column in the table is irrelevant
   v. All entries in a Field/Column must be of the **Same Data Type** e.g. Name, date, number etc.
   vi. No Row/Record in the table can be identical (no repetition)
9. Ms Access

   Ms-Access database consist of a single file that holds the **data** tables, forms, reports, queries and indexes
10. Components/objects of Ms Access database
    i. **Tables** – a table is a database object that consists of Row/Record and Column/Field used to **store** data in a database. It's the basic unit of storage
    ii. **Forms** – is a database object used to simplify data entry, viewing, editing and deleting in a database
    iii. **Reports** – are use for viewing and printing of some or all the data in one or more databases
    iv. **Queries** – are use to select information that meet a given criteria in one or more databases. A select query is used to create subsets of data that you can use to answer specific questions

    Ms-Access provides a "WIZARDS" that help users design queries

- Ms-Access can create a primary key if no single field have the properties required for it (i.e. should be **Unique** and not **Null**)
- Primary key is used to uniquely identify a **Record/Row** in a table
- **Indexing** – an index file contains a subset of the data files based on one or more Key Fields

11. Data integrity
    - Data integrity is the process of ensuring the correctness and completeness of data in the database.
    - This is to ensure validity/validation of data **values** when the contents of the database are modified so that the integrity of the database is not lost
    - Some of the constraints of data integrity are(those that make data have errors):-
      i. Validity integrity – ensure data entered matches the Field/column data type
      ii. Entity integrity – avoid duplication or omission of data/record by ensuring that no primary key is **duplicated** or is left **Null**
      iii. Referential integrity – it establishes a parent-child relationship i.e. any change in the parent table must and should be reflected in the child table.
12. Access interface

    The look and feel of Ms-Access interface is very similar to many of the other Ms-Office products. I.e. it has

Title Bar	Tool Bar	View Bar
Tab Bar	Status Bar	Etc.

13. Database toolbars

    It contains all tools that are needed in access, but are in different Tab in Access 2007 e.g.

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Font formatting tools	Importation and exportation of data to/from external database	
Creation of Tables, Forms, Reports and Queries	Database tools e.g. creation of relationship and analyzing data	

Items found in tool bar

14. Tables

- They are used to store your data in a database. To create a table, you must 1st create a database that will contain all the database objects namely:-

Database objects

- Table can be created either by using the
  - i. Table wizard
  - ii. Design view – used to create the table manually
- Design view has 3 components
  - i. **Field** name – they are user defined and depend on the application to be used for
  - ii. **Data** type – there are different data type in Access e.g. text, memo, number, date and time, currency, yes/no, ole object and auto-number
  - iii. **Description** – use to describe the field

15. Primary key

- A good relational database contains tables where each record/data can be uniquely identified
- One field in the table contains a unique value for each record in the table
- The unique field is called **primary key** and is always recommended that you define a primary key for every table you create

16. Field properties

- It contains two Tabs that can be used to further define the data type and format. They include:-
  - i. **General** Tab - contain most of the information used to define the data type
  - ii. **Lookup** Tab – to look up values from another query or table to supply information to supply information to this field

17. Linking tables

- We link tables to create a relationship
- Relationship is the way in which information in one table is related/associated to information in another table

18. Forms

- It's used to provide a friendly interface between users and Access database
- A Form can be used to:-
  - i. Display and modify data e.g. Add, delete etc.
  - ii. Accept user inputs – accept certain values
  - iii. Controlling application flow
  - iv. Displaying messages e.g. error or warnings
  - v. For printing information

19. Controls tools

- This are tools designed to help users navigate and utilize a Form e.g. Buttons, Radio buttons, text boxes etc.

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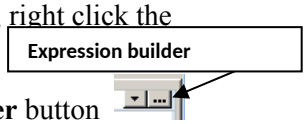
- This controls are broken down into three categories namely:-
  i. **Bound** – this are field of data that come from a table or a query and must have a bound control for each field that you display on the Form.
     Bound control cannot create a calculation
  ii. **Unbound** – contains a label or a text box. Calculation can be created from unbound control
  iii. **Calculated** – are values calculated in the Form e.g. Totals, Subtotals and average

20. Form layout

- All forms must have a Detail section namely:-
  i. **Form Header** – display information that remain the same for every record e.g. Title for the form
  ii. **Page Header** – display information e.g. title or header at the top of every printed page
  iii. **Page Footer** – display information e.g. date or page number at the bottom of every printed page
  iv. **Form Footer** – display information that remain the same for every record e.g. total and sub-total
- You should be able to:-
  i. Create a custom form
  ii. Modify and edit a form e.g. add or remove controls
  iii. Add graphics to a form
  iv. Create a sub-form

21. Expression Builder

- Access includes built-in functions similar to the function in Excel.
- One can also use it to develop owns calculation
- <u>To define an expression, ones need to:-</u>
  i. Select unbound control(or group of instruction) to select, right click the control then –
  ii. Select the **properties** and click on the **expression builder** button



  iii. Begin the expression by typing "=" or click the equal sign button
  iv. Click the button that represent the calculation that you want to enter or double-click on one of the folders that is contained in the left box of the expression builder box
  v. Locate field from table, query, form or report for calculation by double-clicking on the corresponding folder in the left box
  vi. Type any value or constants you would like to include in the expression

22. Querying Data

- A query is a question (or group of instruction) that you ask the database when you want to pull a specific records from the database
- The results/respond is/are set of records that meet the criteria that you set
- A query can also be define as – custom-designed view of data from one or more tables in a database
- In Access, one can query by using:-
  i. SQL statement
  ii. Query wizard
  iii. Designed a query using the **design view**

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- Query can also be **used** to
  i. View data
  ii. Insert data
  iii. Delete data
  iv. Create new table from data retrieved from one or more existing tables
- In SQL view, one need to know how to write queries using the structured query language
- In our class of packages, we will create queries using the design view

23. Common criteria
    i. **Between** *value1* **and** *value2* – find
    ii. **In** – find
    iii. **Null** – find
    iv. **Is not null** – record
    v. **Like** – match criteria
    vi. **Not** =- don't match criteria
24. Wildcard character
    - They are used as placeholders for other characters when you don't know exactly what you are looking for, but you know a part of the value or you want to find values that start/end a latter or match a certain pattern. They include:-
      i. "*" - match any number of characters e.g. wh*
      ii. ? – match any single alphabetical character e.g. b?ll
      iii. [ ] – match any single alphabetical character within the brackets
      iv. ! – match any character not in the brackets e.g. b[ae]ll
      v. " – " – matches any one of a range of characters e.g. d[a–d]d
      vi. # - matches any single numeric character e.g. 1#3
25. Sorting (retrieving records)
    Sorting is – arranging data/record in order, according to a specific field, either in ascending or descending
26. Reports
    - They are the best way to share data with others in a precise and understandable manner
    - Reports are very similar to forms, differences being that you can only use a report to print data and not anything else

One can create a report just like a form

INTRODUCTION TO PRESENTATION

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(MICROSOFT POWERPOINT)

Is a package used to prepare visual aids that can be used in meetings, discussions, forums, seminars e.t.c. or a package used to create designs that can be used for advertising events in form of slides.

Examples of Presentation Packages

These include

- Ms PowerPoint
- Corel presentation
- Soft maker presentation
- Adobe premier

Terminologies

1. ***Slides*** – This are individual screens / pages in a presentation
2. ***Transition*** – Refers to the replacement or wiping effects that PowerPoint uses when one slide is replaced by the next one during a slide show.
3. ***Slide Show*** – Is the final view of the presentation on a full screen where all slides are shown one after the other.
4. ***Animation*** – Is an effect applied to text or graphic so that they look as if in motion.
5. ***Timing*** – This refers to the duration one slide takes on the screen before it's replaced by the next one during a slide show
6. ***Master slide*** – is the first slide within a presentation

Starting PowerPoint

Start → Program → MS PowerPoint

This opens a PowerPoint starter dialog-box which shows the different ways that may be used to create a presentation.

CREATING PRESENTATION

Presentation may be created in four (4) different ways.

a. AutoContent Wizard
b. Design Template
c. Blank Presentation
d. From an Existing File

<u>Auto content Wizard</u>

- This enables the user to create a new presentation out of sample lines already created and saved in the computer.
- That is a kind of a template that gives guidelines and design that may be used to create a new presentation and it is done by way of replacing the information on slides with your new ideas that will form the new presentation.

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To use AutoContent Wizard,

- Start PowerPoint & click on the ***Office Button***
- Click on ***New*** from the list of commands, a ***New Presentation*** dialog box will appear
- Under the ***Templates*** option, click on ***Installed templates*** Command
- Select a theme of your choice from the list that appears then click ***Create***

Design Template

- This the one used to design a presentation using different background provided by PowerPoint.
- The selected background is applied on the slide known as the Master Slide.
- This method is commonly used to create a one-slide presentation such as a poster or an advertisement and when printed, it may appear like a Word Processing document.

To do this

- ✓ Start PowerPoint & click on the ***Office Button***
- ✓ Click on ***New*** from the list of commands, a ***New Presentation*** dialog box will appear
- ✓ Under the ***Templates*** option, click on ***Installed themes*** Command
- ✓ Select a theme of your choice from the list that appears then click ***Create***

NB: Instructions include click to add text / click to add sub-title etc

From An Existing Presentation

- This enables one to create a presentation in cases where identical presentations need be.
- Layout and design of existing presentation are used for a new one.

To do this

- ✓ Start PowerPoint & click on the ***Office Button***
- ✓ Click on ***New*** from the list of commands, a ***New Presentation*** dialog box will appear
- ✓ Under the ***Templates*** option, click on ***New from existing*** Command
- ✓ A ***New from existing presentation*** dialog box will appear.
- ✓ Select / choose the location of the saved presentation that you want to use then click on ***Create new***
- ✓ Replace the information on the new slide with new information i.e. your own ideas

Blank Presentation

This allows the user to create a presentation from scratch, i.e. every element of the presentation (ideas, layout, text, graphics…) is determined by the user.

To do this

- ✓ Start PowerPoint & click on the ***Office Button***

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- ✓ Click on ***New*** from the list of commands, a ***New Presentation*** dialog box will appear
- ✓ Under the ***Templates*** option, click on ***Blank and recent*** Command
- ✓ Select ***Blank presentation*** from the list that appears then click ***Create***

Saving a Presentation

- ✓ Click on ***Office button*** to display a list of commands
- ✓ Move to **"*Save As*"** option to obtain a sub menu
- ✓ From the sub menu click on ***PowerPoint 97_2003 presentation*** option
- ✓ A save as dialog box will be displayed
- ✓ Type the ***File Name*** in the File Name box
- ✓ Select the name of the Folder / Drive in which to save the file
- ✓ Select your favored destination then click ***Save***.

Viewing Presentation

A PowerPoint Presentation can be viewed in different ways depending on the operation carried out. That is

1. Normal View
2. Slide Sorter View
3. Notes Page View
4. Slide Show View

Normal View

- This view is normally very useful during the initial stages of creating a presentation.
- One can quickly move to a particular slide by clicking on its slide number from the left panel for the actual slide to be displayed on the right panel.

To do this

- ✓ Click on ***View tab***
- ✓ Click on the ***Normal*** command under the ***Presentation views*** group

Or

- ✓ Click on the ***Normal*** command after the ***Status bar***

Slide Sorter View

- Is a view that displays all the slides in the presentation on one screen.
- It acts as the Print Preview to the presentation.
- It is very important when we wish to add / delete / remove slides

To do this

- ✓ Click on ***View tab***
- ✓ Click on the ***Slide sorter*** command under the ***Presentation views*** group

Or

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- ✓ Click on the ***Slide Sorter*** command after the ***Status bar***

Notes Page View

- This view divides the screen into 2 parts – the slide at the top and the notes option at the bottom.
- The notes part displays the flow of ideas while that on the top displays the actual slide being worked on.
- The notes page view is used to edit the speaker notes as they will look when you print them out.

To do this

- ✓ Click on ***View tab***
- ✓ Click on the ***Notes Page*** command under the ***Presentation views*** group

Slide Show View

- Is the final view of a presentation that is shown to the audience where slides are shown one after the other on full screen.
- The slides may be made to 'transmit' automatically or manually by the mouse

To do this

- ✓ Click on ***View tab***
- ✓ Click on the ***Slide show*** command under the ***Presentation views*** group

Or

- ✓ Press F5

Or

- ✓ Click on ***Slide Show tab***
- ✓ Click on the ***From beginning*** command under the ***Start Slide Show*** group

Or

- ✓ Click on the ***Slide show*** command after the ***Status bar***

Switching Between Views

There are two ways to do this

i. Go to ***View tab*** then click on the suitable view under the ***Presentation views*** group.

ii. Click on the ***viewing commands*** on the ***right hand side*** of the ***status bar***

Inserting Slides

After creating the first slide of a presentation (the master slide), most slides may be inserted to complete the show.

- ✓ On the ***Home tab,*** click on arrow pointing downwards on the ***New slide*** command under the ***Slides group***
- ✓ Select the slide layout of your choice

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- ✓ The new slide will automatically be inserted in your presentation.
- ✓ Enter text in your new slide

Or

- ✓ Press Ctrl + M on the keyboard

Or

- ✓ Right click on the last slide then from the shortcut menu click on ***New slide*** command

Slide Layout

- This refers to how you want your information to be arranged on your slide.
- There are different types of slide layout depending on the type of information or data to be entered in a slide. A few examples of different layouts are :-

a. Inserting Tabular Information

This is used when information has to be presented in tables for clarity e.g. a price list

To do this

- ✓ Insert a new slide
- ✓ Select a layout that has a table
- ✓ Click on the ***Insert table*** command on the slide
- ✓ An ***insert table*** dialog box will appear
- ✓ Specify the number of columns and rows that you want then click ***OK***

b. Graphical Information

Helps in presenting information using graphs and charts e.g. comparison of some

products e.t.c

To do this

- ✓ Insert a new slide
- ✓ Select a layout that has a ***chart***
- ✓ Click on the **Insert chart** command on the slide
- ✓ An insert chart dialog box will appear, select the ***Chart type*** and ***chart sub-type*** then click ***OK***
- ✓ A sample table will open in Ms. Excel, fill the table with your own information which will be reflected in the chart which is in Ms, PowerPoint
- ✓ Close the Ms. Excel windows once through entering your data. The changes made will be reflected in your chart

NB – you can format the chart the same as in Ms. Excel by activating it and the Chart tools tab will appear

c. Smart Art Graphic

Helps in presenting information in form of ***a list***, ***Process***, ***Cycle,*** H***ierarchy***, ***Relationship,***

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Matrix and Pyramid

To do this

- ✓ Insert a new slide
- ✓ Select a layout that has ***SmartArt Graphic***
- ✓ Click on the ***Insert SmartArt Graphic*** command on the slide
- ✓ A ***Choose a SmartArt Graphic*** dialog box will appear where you choose the graphic you want i.e. from the ones listed above
- ✓ Click ***OK***
- ✓ Enter your data

d. Inserting Picture from file

This is used when you want to insert a picture that you have saved in your computer

To do this

- ✓ Insert a new slide
- ✓ Select a layout that has ***Picture from file***
- ✓ Click on the ***Inserting Picture from file*** command on the slide
- ✓ An ***Insert Picture*** dialog box will appear where you select the location of your saved picture
- ✓ Select the picture you want to insert then click on ***Insert*** command
- ✓ The picture will automatically be inserted in your slide
- ✓ Move your mouse pointer on the square/round handles of the picture, your pointer will assume the shape of a double headed arrow
- ✓ Click and drag your mouse to increase / decrease the size of the picture

e. Inserting Clip Arts

Clip arts are used to make slides more attractive as well as easily understood.

To do this

- ✓ Insert a new slide
- ✓ Select a layout that has a clip art
- ✓ Click on the ***clip art*** command on the slide
- ✓ A ***clip art*** dialog box will appear. Type the category of clip art you want to insert then click ***Go***
- ✓ Select the suitable clip art / picture from the art gallery and will automatically be inserted in your slide
- ✓ Move your mouse pointer on the square/round handles of the picture, your pointer will assume the shape of a double headed arrow
- ✓ Click and drag your mouse to increase / decrease the size of the picture

f. Inserting Media Clip

This is used when you want to insert a short video / media clip in your presentation

- ✓ Insert a new slide
- ✓ Select a layout that has ***Media Clip***
- ✓ Click on the ***Inserting Media Clip*** command on the slide

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- ✓ An ***Insert movie*** dialog box will appear where you select the location of your saved movie
- ✓ Select the media clip / movie then click ***OK***

Deleting Slides

Is the removal of unwanted slides from a presentation.

When deleted, a slide is completely removed with all its contents

- Go to ***View*** tab
- Select ***Slide Sorter View*** under the ***Presentation views*** group
- Select the slide to be deleted by clicking on it once
- Press Delete key on the keyboard ***or Right click*** on the slide then click ***Delete slide***

Or

- Right click on the slide you want to delete in ***Normal View***
- From the shortcut menu click ***Delete slide***

Or

- Select the slide you want to delete
- Go to ***Home*** tab
- Click on ***Delete*** command under the ***Slides*** group

Running a Slide Show

- A Slide Show is usually the final viewing of a presentation that is shown on the full screen where all the info in all slides is displayed in a sequence by which they are show, corresponding with the flow of ideas as set by the user.
- A slide show may be run either-
  a. Manually

  b. Automatically

Manually

A slide show is said to be running manually where a mouse click helps to move from one slide to another i.e. the user runs the show by clicking on the mouse

To do this

- ✓ Click on ***View tab***
- ✓ Click on the ***Slide show*** command under the ***Presentation views*** group
- ✓ Click the left mouse button or press the left arrow key until the end of the show.

Or

- ✓ Press F5
- ✓ Click the left mouse button or press the left arrow key until the end of the show.

Or

- ✓ Click on ***Slide Show tab***
- ✓ Click on the ***From beginning*** command under the ***Start Slide Show*** group

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- ✓ Click the left mouse button or press the left arrow key until the end of the show.

Or

- ✓ Click on the ***Slide show*** command after the ***Status bar***
- Click the left mouse button or press the left arrow key until the end of the show.

Automatically

- Is the running of the show from the 1st slide to the last unassisted.
- This will involve initial setting, timings and transitions in the slides.

To do this

- ✓ Click on ***View tab***
- ✓ Click on the ***Slide show*** command under the ***Presentation views*** group
- ✓ Let the slide show run without clicking on the mouse button or using the arrow key.

Or

- ✓ Press F5
- ✓ Let the slide show run without clicking on the mouse button or using the arrow key.

Or

- ✓ Click on ***Slide Show tab***
- ✓ Click on the ***From beginning*** command under the ***Start Slide Show*** group
- ✓ Let the slide show run without clicking on the mouse button or using the arrow key.

Or

- ✓ Click on the ***Slide show*** command after the ***Status bar***
- ✓ Let the slide show run without clicking on the mouse button or using the arrow key.

Setting a Transition

This shows a professional progression of slides whereby each slide will appear and disappear automatically in a set style.

To do this

- ✓ Go to ***View*** tab
- ✓ Select ***Slide Sorter View*** under the ***Presentation views*** group
- ✓ While on Slide sorter view, click on the ***Animations*** tab
- ✓ Select the first slide then select the suitable transition under the ***Transition to this slide*** group
- ✓ Select the transition ***sound*** i.e. click on the arrow pointing downwards and select a sound from the list
- ✓ Select the ***speed*** i.e. slow, medium or fast by clicking on the down arrow
- ✓ Move to the ***Advance slide*** part and activate the ***Automatically after*** checkbox then set the time by clicking on the spin button
- ✓ Select the next slide and apply the above settings
- ✓ Go to ***Slide show*** tab, click on ***Set up slide show*** command under the ***Set up*** group
- ✓ A ***Set up show*** dialog box will appear, click on ***Using timings if present*** command under ***Advance slide*** options
- ✓ Click ***OK***

Setting Animation Effects

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- Animation is that effect set where the slides appear as if in motion.
- Effects are applied to objects, texts and graphs in order to create suspense and/or arouse interest in the viewers
- It is applied per object per slide
- When setting animation, each slide takes animation which is set separately, i.e. to set animation on 'Slide 1' Select Slide 1 → Repeat for other slides

To do this

- ✓ Display the Slides in ***Normal*** View then click on the slide that you want to work on
- ✓ Select the ***Object / Part*** you want to work on i.e. title, subtitle, text, clip art e.t.c
- ✓ Click on the ***Animations*** tab then click on ***Custom animation*** under ***animations group***
- ✓ A ***Custom animation dialog box*** will appear, click on the ***add effect*** command then from the options i.e. ***Entrance, Emphasis*** and ***Exit*** select the suitable animation effect(s)
- ✓ Repeat the above procedure to all the other objects in the slide
- ✓ Click OK once done
- ✓ Select the next slide then repeat the above procedure(s) again.

Close the custom animation dialog box once you are through and run the show

6. Internet and E-mails (Computer Networking)

1. Definition

- **Network**
  - It's a set of **connected** things or people and its' size is determined by the **number** of items/people connected to it, their **locations/area size** (space between them), the **needs** of the users (capabilities/data band) and the available **technology** e.g. inter-county highways, telephone network etc.
- **Computer network**
  - It's formed when two or more computers are connected together to enable **communication** among them e.g. send messages, share files and programs
- **Internet**
  - It's the world biggest computer network (a vast network of networks)

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- Internet comprises of the **hardware** that sent information/data from one place to another and the **software** that govern (glues) the system together

- Network that form the internet are **individually** run by different entities namely the: - government agencies, universities, commercial and voluntary organization etc.
- **No** single organization owns or controls the internet, though there is an internet society (The World Wide Web Consortium – W3C) that set and co-ordinates standards and protocols for the internet.

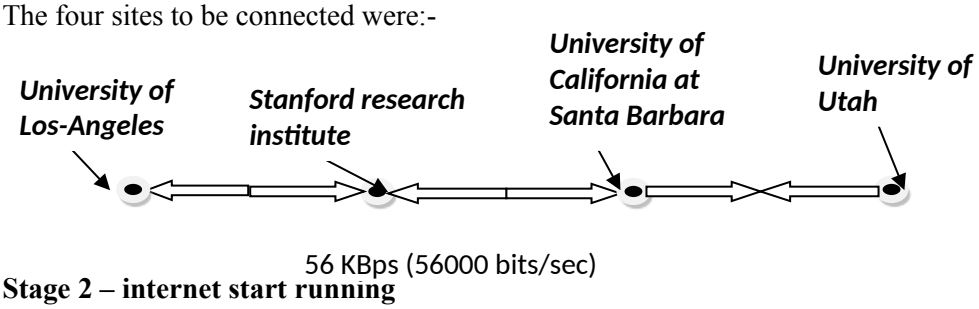
2. History of the internet

- **Stage 1 – The beginning**

  In 1969, the Defense Advance Research Project Agency (DARPA) of the USA Department of Defense (DoD) built the 1st network known as Advance Research Project Agency Network (ARPANET)

  The intention was to built a network that could continue to work even if part of the network were disabled/destroyed

  The four sites to be connected were:-



- **Stage 2 – internet start running**

  In 1982, the Defense Communication Agency (**DCA**) and **DARPA** established a protocol known as **Transmission Control Protocol and Internet Protocol (TCP/IP)** as the standard protocol used on the network (ARPANET) and the conversion to TCP/IP on the ARPANET was completed on jun. 1983 and the term "INTERNET" come into use in place of the "Internet Protocol (IP)"

  In 1989, ARPANET was split into two networks namely

  - MILNET – military network for military department
  - ARPANET – for non-DOD sites and the TCP/IP was the accepted protocol running on the network

  In 1984, Japan Started **JUNET** (Japan UNIX Network) and UK started **JANET** (Joint Academic and Research Network)

- **Stage 3 – national science foundation**

  In 1986, the DOD assigned the ARPANET project to the National Science Foundation (NSF) and a network connecting five supercomputers centers was established by the NSF to enhance computer and network research capabilities in the USA

  The NFS network and other network makes up the largest network of networks; the Internet

- **Stage 4 – the World Wide Web (WWW)**
  - Event that began in 1989 and became public in 1992 was the 1st public distribution of the World Wide Web (WWW) **client Software** on the Internet
  - The WWW client software was created by **Tim Berners** at the Conseil Europeen por la researche Nucleare (CERN) – (European Consul for Nuclear Research) laboratory of particle physic. He propose a **hypertext**

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system enabling scientist studying high-energy physics to share information more efficiently

- The proposal was very important coz it involve a user **interface** that would be consistent across all platform e.g. UNIX, Macintosh, Windows etc.
- The WWW uses the Internet Protocol **http:/** to present information through multimedia format; ie graphics, sound, animation and video
- Internet (WWW) resemble an enormous **electronic library**; each "**Web Site**" is comparable to a **book**; each book as a **page(web page)** which contains content and links to other pages(**Web Page**) in the book(**Web Site**) and also other books (**Sites**)
- WWW client **programs/software** include :-

Internet explorer	Netscape	Mosaic
Mozilla Firefox	Lynx	Charlotte
Opera mini	Ect.	

3. Advantages of the internet

i. **Speedy** communication
ii. **inexpensive** communication
iii. It's provide **convenient** communication

4. Uses of internet

- **Communication services:- e.g.**
  i. E-mail
  ii. Chatting
  iii. Conferencing
- **Commerce and industries**

  It facilitate trade and is replacing traditional practices e.g. Tel, Fax, Letters

  It also enables
  i. E – business;
  ii. E – commerce
     1. Online shopping
     2. Online banking

  Some advantages of internet on commerce are:-
  i. Ability to advertise and sell to the world
  ii. Fast and cheaper communication with suppliers and consumer vie e-mail
  iii. Creating seamless online supply – distribution chain for seekers and customers
- **Health**

  It provides information, daily news and other resources about health.
- **News and information**
  - One can get headlines information and in-depth stories from the latest political going on, to games to the hottest technology available
  - It also provide information on weather around the globe as well as travel information and tourist guides
- **Job opportunity**

  With the use of internet, one can find an ideal job
- **Education**

  They include:-

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i. Virtual learning
ii. Reading book and magazines in the internet
iii. In-service training for teachers and lectures

- **Software download**
  One can download and use a program from the internet
- **Finding people and organization**
  One can find people (e.g. using face-book) or an organization (e.g. a school) by using the internet
- **Linking to traditional communication media**
  With internet, one can access

Telephone	Mobile phone	Short Message Service(SMS)
Banking	E-mails	Radio/Tv

5. Internet connection

The following elements are required for one to access the internet

i. **Computer**
   Maybe

ii. **Modem**
   - A modem is device that form a connection, (via telephone or cable-TV lines) ,between your computer & a computer network that provides internet access.
   - Modem may be internal or external & the wires that carry information are plugged into your modem.
   - Modems require software that will enable you to control your modem.

iii. **Internet connection**
   Internet connection is classified into two methods namely:-

   a. Dial-up connection
   - It accesses the internet on a per-use basis & uses a modem.
   - It gains internet access when they contact an ISP & the ISP transfers them to the internet. When finish they disconnect from the ISP. The speed depends on the modem.
   - To gain faster dial-up connections, you can install an Integrated Services Digital Network (ISDN) line, which is a digital tel. line.
   - In an ISDN modem, a "terminal adapter" is attached to the computer that connects to the internet. The ISDN lines are expensive way to access internet in a dial-up connection.
   - The most inexpensive method of accessing Internet is by using a telephone line, an analog modem and a subscription to an ISP

   b. Direct connection
   Its Provide continuous, "Always on" access to the internet
   It's convenient and fast because of high band-width
   Direct connection includes:-
   i. Local Area Network(LAN) connection (using a router)
   ii. Cable TV (using Network Interface Card(NIC) or cable box)
   iii. Digital Subscriber Line(DSL) – by Digital phone lines, a DSL modem and a NIC to attach to the computer

   Direct connection are commonly referred to as **Broadband** connection

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c. Factors to consider when selecting a good Internet Service Provider(ISP)

- Internet Service Providers (ISP) are organizations /or/companies that provide individuals and businesses with access to the Internet through Dial-up or Broadband connection.
- When choosing an internet service provider (ISP), one should consider the following factors

Ease of use	Support(customer service)	Cost
Services offered e.g. E-mail	Trial before buying	Content offered
Performance(speed)	Privacy	security

iv. Internet software(Web-Browser)

Communication software is needed to manage your connection to the internet.
Some of the application programs (client software) available include

Internet Explorer	Opera Mini	Mozilla Firefox
Netscape		

6. Accessing the internet

- **Login/sign-in**
  ISP will provide you with a **username** and a **password**
- **Getting online**
  - One has to dial to the ISP and may require a Username and a Password to gain access to the ISP network
  - Some OS (Operating System) or Wed browsers may give an option to remember password

7. What makes up the Web(Internet)

Internet is made-up of 3 components namely:-

i. Web site

It's made up of collection of web pages which may include anywhere from a handful up to millions of web pages, published as one interconnected entity e.g. Wikipedia, Google, Nation-media, BBC etc.

ii. Web pages

It's a single page in a website. Ti provides a single screen of information which may contain text, images, animation, sound, video etc.
You can view a web-page via the web-browser software

iii. Links

These are Hypertext or Hypermedia that is used to move from one page to another within a website or to another webpage.

iv. Hypertext

- Hypertext is text like regular text (it can be stored, read, search, edited, etc.) with **one exception**: - "hypertext" contains within it **a connection** to other sections/text or documents, /or/
- Hypertext is a structured text that uses logical links (hyperlink) between **nodes** containing text

v. Hyperlink (Hypertext links/connection)

This are connection that are used to create complex virtual web of connections
Hyperlinks are usually underlined and blue in color. When you click it, its color changes to red

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vi. **Hypermedia**

It contains not only to other pieces of text, but also to other forms of media i.e. sound, images and movies. Images can be selected to link to sound or documents

8. How the Web Works

- **Hypertext transmission protocol (HTTP)**
  - This is the protocol that Web-client and servers computers use to communicate with each other (it's the protocol used to exchange or transfer text)
  - The web works under the popular client-server model
  - All web client and servers must be able to speak HTTP in order to send and receive hypermedia documents e.g.
    - i. Web servers are often called HTTP servers
    - ii. Web addresses are automatically prefixed with **http:/**
  - The phrase "**World Wide Web-(WWW)**" is often used to refer to the collection of network of computers and servers that are speaking **HTML language** and sharing information using the **protocol HTTP**
- **Hypertext markup language (HTML)**
  - This is the language the web uses to creating and recognizing hypermedia documents
  - HTML is widely praised for its ease of use. Documents are usually named with the suffix ".html"

9. Uniform resource locator (URL)

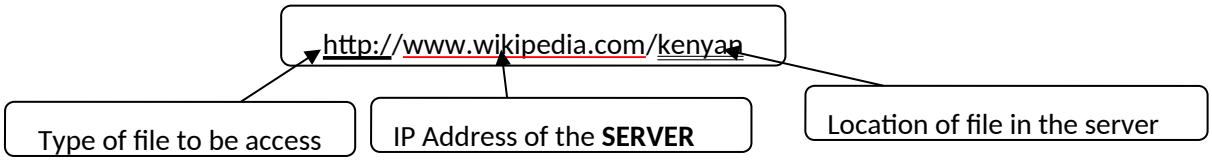
URL is the Internet address of a particular Web Page (acts like a tel. number)

- **Structure of URL**

  URL address has two basic parts namely:-

  i. A portion specifying the method of **file access**
     - The 1st part that end with "://" tell the type of file that is permitted from the internet
     - It tells the **server** computer which **type** of file the **client** computer is requesting
     - e.g. http:// stands for "Hypertext Transfer Protocol" and tells the **server** that the type of file you have requested

  ii. A portion specifying the **internet location** of the file to be accessed
     - It specifies the internet address of the machine housing the file and the location of the file at that internet address.
     - The rest of the URL address after "**.com/**" gives the location of the file on the computer



10. Surfing/Browsing the internet

- **Starting up your browser**
  - You start up your browser in the same way as any other software e.g. Ms Word, Ms excel. However, you need to be connected to the **Internet** 1st to access the web page
  - Some computers may be configured to automatically connect to the Internet upon opening the web browser

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- Web browser menu is the same as that of any other Windows Program

- **Requesting a web page**
  - There are several ways of requesting a web page. They includes:-
    i. Type a "**URL**" in the **address bar** on the web browser and click "**GO**" or hit **"ENTER"**
    ii. Click "**file**" menu, select "**open page**" and key in the URL
    iii. Press "**Ctrl + D**" at the same time, then type in the URL you want

11. Searching the web (Using Search Engines)

- Search engines allow the users to type keywords that are run against the database on the server computer
- It uses the software "**reboot or spiders"** to **crawl** the web and build its database. Web documents are retrieved and indexed
- When you enter a query at the search engine web site, your input is checked against the search engine "**keyword**" indexes.
- When using the internet, you cannot search the internet **live,** but you are searching a pre-compile database
- While all **search** engine are intended to perform the same task, each engine does it differently, which sometime leads to significantly different results
- Factors that influence results include:-
  i. Size of the database
  ii. Frequency of update
  iii. Search capabilities e.g. search speed, design interface, help they offer
- Search engines are best used to locate a specific piece of information, e.g. document, image, program etc. rather than a general subject
- The growth of the number of search engine has led to the creation of "META" search tools, often referred to as "**Multi-threaded-search engines**". These search engine allow the user to search multiple databases simultaneously, via a single interface
- Example of search engine are:-

Google	Mozilla	MSN
Yahoo	HotBot	Lycos
Dogpiles	Excite	AltaAista

12. Common surfing problems

- When using web, common problems includes:-
  i. URL not working due to typing errors
  ii. Dead links
  iii. Slow in loading content
  iv. Telecommunication breakdown
- Other problems which are associated with URL error and is given in error code message includes

400 – bad request	401 – unauthorized	403 – forbidden
404 – file not found	500 – server error	503 – server unavailable
Server under construction	Server not responding	Etc.

13. Web directories

- This is hierarchical organized **Indexes** of subjects (files) categories, that allow the web searcher to browse through a list of web sites by subject in search of relevant information

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- They are compiled and maintained by humans and may include a search engine for searching their own database.
- **Web directories** greatly reduce the probability of retrieving results out of context coz it include human intervention
- Web directories are usually smaller than those of the search Engines hence the result list tend to be small as well
- Examples of subject directories includes:-
  i. Look smart (www.looksmart.com)
  ii. Open directories (www.dmoz.com)
  iii. Yahoo (www.yahoo.com)

14. Methods of text indexing

- There are two primary methods of text indexing, namely:-

a. Keyword Indexing

- This is the most common form of text indexing on the web.
- In this method, text are indexed by use of **key word** or in some case, by **phrases**
- Unless the author specifies the key-word for the document, it's up to the search engine to determined them by pulling out words that it believes to be of significance e.g. words at the top of the document or those that are repeated more often in throughout the document
- Key word **searches** find it difficult to distinguish between words that are **spelled** the **same way** but mean **different things** /OR/ words spelled **differentl**y but mean **the same thing**; therefore may results in hits that are completely irrelevant to the query
- It cannot return hits on key-words that mean the same thing but are not actually entered in your query e.g. a query on heart diseases would not return a document that use the word "cardiac" instead of "heart"

b. Concept-base Indexing

- It tries to determine what you mean(unlike key-word indexing), not just what you say
- They do this by checking documents for the dominant theme or concept, which are then indexed.
- They return hits on documents that are about the subject/theme you are exploring, even if the words in the document do not precisely match the word you entered into the query
- Some concept base indexing rely on sophisticated linguistics and artificial intelligence theory e.g. by calculating the frequency with which certain important words appears. The search engine concludes, by statistical analysis, that the pieces are "about" a certain subjects e.g. the word "heart", when used in the **medical/health** context, would be likely to appear with words such as coronary, artery, lung, stroke, cholesterol, blood, arteriosclerosis etc. if the same word "hearth" appears in a document with other words such as flower, candy, love, passion, valentine etc., a very different context is established and the search engine returns hits on the subject of **romance**

15. Electronic mails (E-mail)

It's a type of mail that is **electronic,** whether it's **transmitted** over the internet or not.

a. E-mail software

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- E-mail allows one to compose, send and receive e-mails. The most e-mail software includes: - Microsoft Outlook, Netscape communicator, Eudora etc.
- The e-mails are stored in a separate folders described below

Inbox	Store incoming e-mails. It's also store older emails or easy use
Outbox/sent	Contain sent messages
Draft	Emails that are being worked on and not yet ready to be sent
Deleted items	Emails that have been deleted

b. <u>Composing an e-mail</u>

- This is the process of writing an email.
- It's best to write when the internet is not connected(off-line)
- Most header **<u>elements</u>** in the new e-mail window are
  i. **To** – address of your recipient
  ii. **Subject** – a few words outlining the nature of your message
  iii. **CC** – copy an email to other recipient
  iv. **BCC** – copy an email to other recipient without any indication of that action appearing in the main recipient's copy of the message
  v. **Attachments** – to attach a file to your email

c. <u>Sending, receiving and forwarding an E-mail</u>

- Once one has finish composing an email, he/she can send it immediately or save it as draft so that he/she can send it later
- Once an email have send an email, the send massage automatically appears in the "**sent items/outbox**"

d. <u>Attaching documents to E-mail</u>

- One can attach a file, document, picture, sound or a movie to an email before sending it.
- All email software provide tools for attaching a document in an email and will send the attached document automatically when the email is sent

e. <u>Deleting mails</u>

- One can delete an email message from the **Inbox** folder when need be and the deleted message is moved automatically to ht Trash/Deleted items

f. <u>Mail Etiquette</u>

When writing an email, the following measures should be taken into consideration:-

i. Keep emails short and clear
ii. Avoid forwarding other people massages

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7. Data security

1. Definition

- Data security is the protection of computer hardware, computer software (programs) and data in the computer and communication system against unauthorized modification, destruction, disclosure or transfer whether accidental or intentional

2. Data control

- Data control is the measures taken to enforce the security of the programs and data
- In order to protect against data lose, control measures need to be put in place. These controls may be just procedures, software or hardware
- Usually a combinations of all will be required to protect against all possible threats

3. Various threats to data security.

i. Physical damage and theft

i. Theft

- In an event of computer theft, one will lose data and information stored in it plus computer itself
- In most of the cases, data stored in the computer is by far more expensive that the computer itself.

ii. System crash

This is the total computer failure that occurs when the computer is damaged.

iii. Protection against theft and system crash

One should regularly backup his/her data so that you can restore the information to the point of last back up. Incomplete backup (infrequent, incomplete), may lead to

a. Loss of data stored in the system
b. In the case of computer theft, one will loss the hardware itself

ii. Viruses

i. Definition

- This is a hidden self-replicating computer program/software written with ill motives by mischievous individuals with bad intention and spread to other computers by sharing infected files or/and programs
- Viruses are designed to "**self replicate**" or "**clone**" themselves, that is, secretly copy themselves from place to place. This process is called infection

ii. Virus detection – we detect a virus when we experience one or more of the following symptoms

- Slow down of computer programs or slow down of the computer general performance
- Unusual and frequent error messages
- Change in file size or replication of computer files
- Loss or modification of data
- System failure

iii. Protection against viruses

- Installing Anti-virus software in your computer
- Files should be backup so that it can be restored back in the event of a system failure
- One should avoid using programs of unknown origins

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- ❖ One should not open an attached file from unknown origin in an email
- ❖ Always check for viruses on portable disks when using it to transfer files from one computer to another

iii. **Software Bugs**

- ❖ These are errors in computer software that are usually unintentional.
- ❖ Running a computer with a bug can lead to serious malfunctioning to the computer and may lead to loss of data

iv. **Trojan horses**

- ❖ These are dangerous pieces of computer program that comes intentionally hidden or appears like a software that is well-behave and useful

v. **Software chameleons**

- ❖ This type of program is made to look harmless but in the background, it carries out illegal adjustments to your computer.
- ❖ E.g. it may be designed to record your secret password or steal a thiny amount of money over a long period of time.

vi. **Software bombs**

- ❖ This is programs which cause destruction as soon as you try to execute or run it.

vii. **Logical bombs**

- ❖ Works like software bombs but it triggered by some condition such as an attempt to carry out a certain operation such as saving a file.

viii. **Time bombs**

- ❖ These are programs triggered by dates or after a certain number of a particular operations has been carried out.(e.g. on 12th dec 1999 live Mellissa)

ix. **Replicators**

- ❖ This copies itself onto the disk &memory (ram) so many times that space is not available for other programs to run thus crippling the computer.

x. **Worms**

- ❖ These travel across networks & are usually designed to cause alteration similar to software chameleons wherever they go.
- ❖ They are difficult to detect with antivirus.

4. Computer crimes

i. **Hacking**

- ❖ This is when an individual gets unauthorized access into the computer system
- ❖ Data being transmitted in a network can also be intercepted
- ❖ Once an hacker breaks into the system, they can
  - **i.** Take a look at the data and information stored in the system
  - **ii.** Steal the information
  - **iii.** Modify or destroy the information
- ❖ Protection against hacking
  - **i.** Log off or shut down the computer when one is not using it
  - ii. Choose passwords that are not obvious and keep them confidential and should not be written down anywhere
  - iii. Change password frequently

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iv. Encrypt data before sending them over the network
v. One should also use other methods of authentication e.g. fingerprints, voice recognition, face recognition etc.

ii. **Trespass**
This is when a person is able to connect to your computer through a computer network and is able to control it in some way, e.g. accessing a file or running a program.

iii. **Tracking**
This is when a person can monitor what you are doing on your computer or your current location from a remote computer.

iv. **Tapping**
This is when a person can gain access to information that is being transmitted via communication channel.

v. **Cracking**
This is similar to hacking only that in cracking, one is doing so with the intention of causing damage or self gain

vi. **Sabotage**
This is when a person renders another person's system unusable through a number of activities e.g. denial of service

vii. **Spam**
This is when a person sends unsolicited junk email, called spam, by accessing a list of email address, to thousands of users

5. Privacy and fraud

- Privacy is when a person is able to keep his/her secrets without other people knowing them.
- Cracking, trespassing, tracking, tapping and hacking are all a violation of one's privacy and are generally considered an offence/crime/fraud in most countries

6. Detection and protection of computer crimes

One should take the following precautions

i. **Physical security**
ii. **Audit trail**
iii. **Data encryption**
iv. **Firewalls**
v. **install antivirus**

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